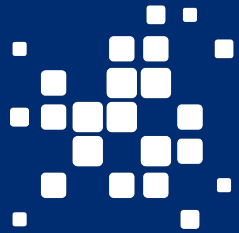
Pangenome case studies

Presented by Katie

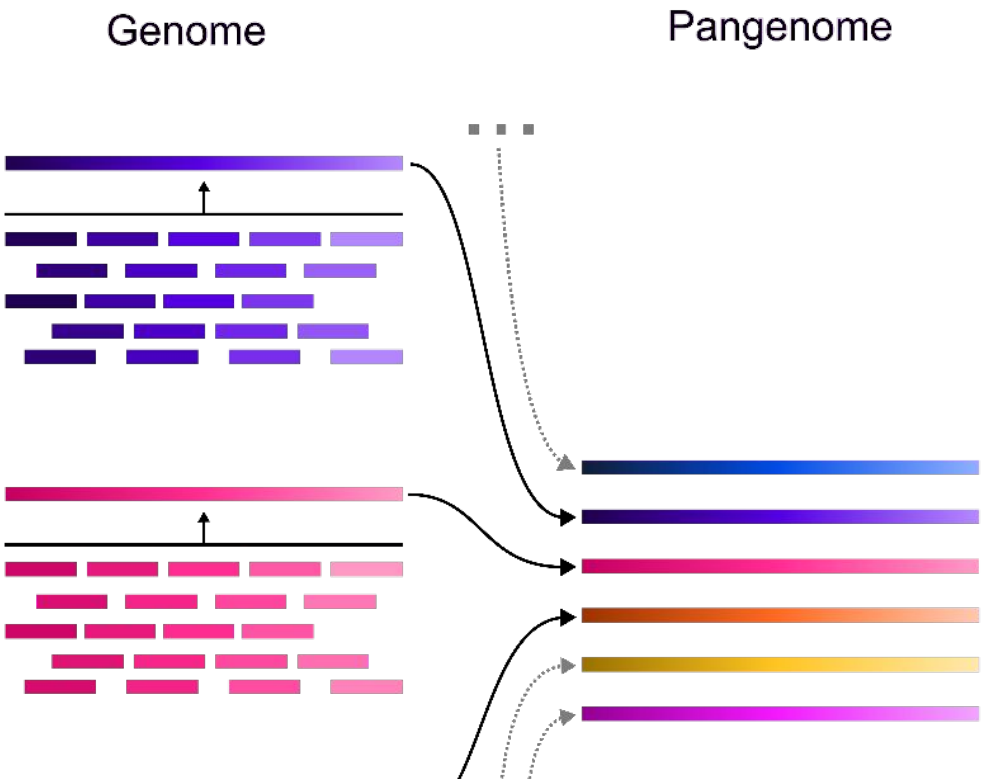


wellcome sanger institute

Outline: Examples of how these tools have been used and combined to answer interesting questions

Genome QC

Genome scaffolding

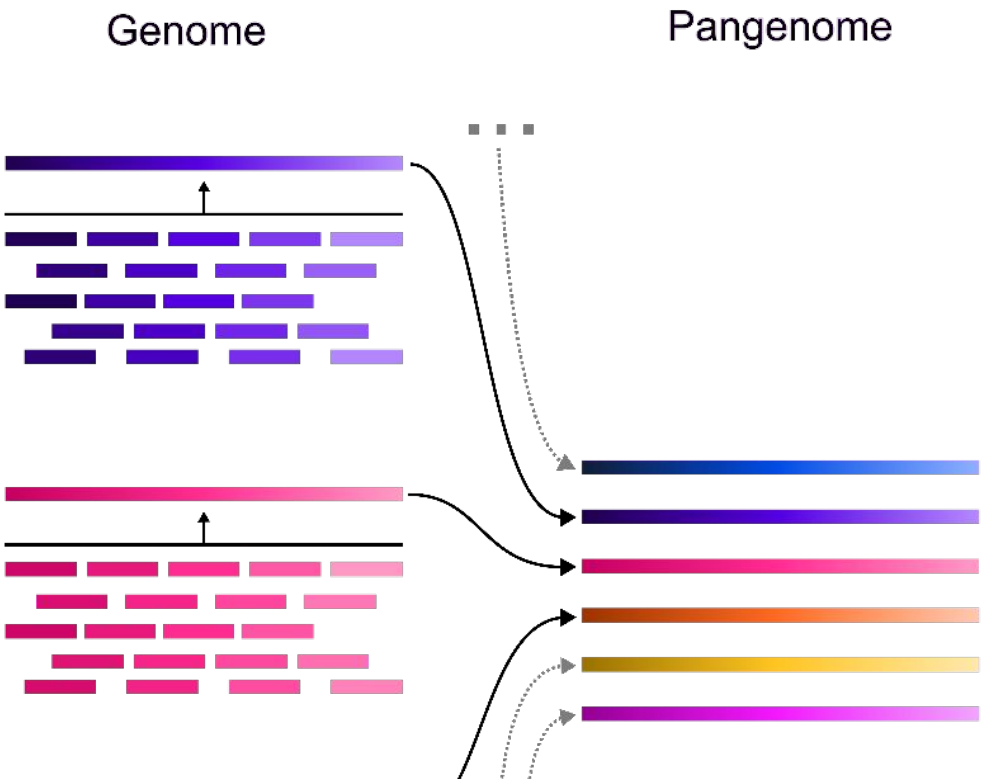


Examples of pangenomes in the wild

Outline: Examples of how these tools have been used and combined to answer interesting questions

Genome QC

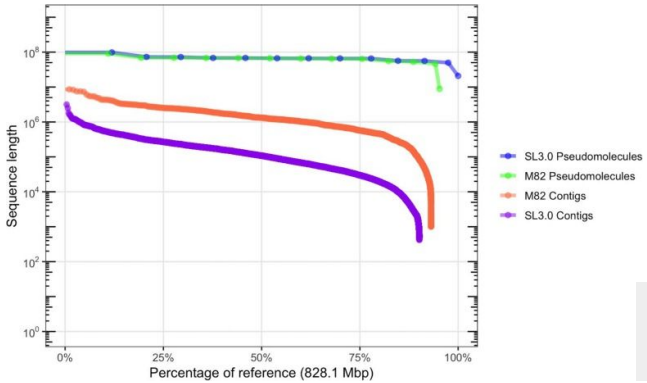
Genome scaffolding



Examples of pangenomes in the wild

Genome assembly quality control

Nchart + associated stats



Merqury



Genome Biol. 2020 Sep 14;21:245. doi: 10.1186/s13059-020-02134-9

Merqury: reference-free quality, completeness, and phasing assessment for genome assemblies

Arang Rhie [1,✉], Brian P Walenz [1], Sergey Koren [1], Adam M Phillippy [1]

Author information · Article notes · Copyright and License information

PMCID: PMC7488777 PMID: 32928274

BUSCO completeness

JOURNAL ARTICLE

BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs FREE

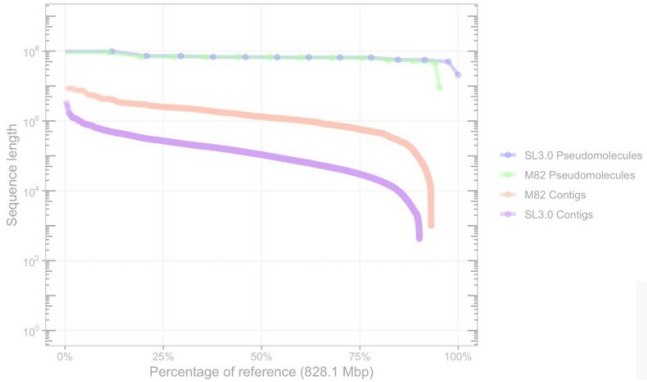
Felipe A. Simão, Robert M. Waterhouse, Panagiotis Ioannidis, Evgenia V. Kriventseva, Evgeny M. Zdobnov ✉ Author Notes

Bioinformatics, Volume 31, Issue 19, October 2015, Pages 3210–3212, https://doi.org/10.1093/bioinformatics/btv351

Published: 09 June 2015 **Article history** ▾

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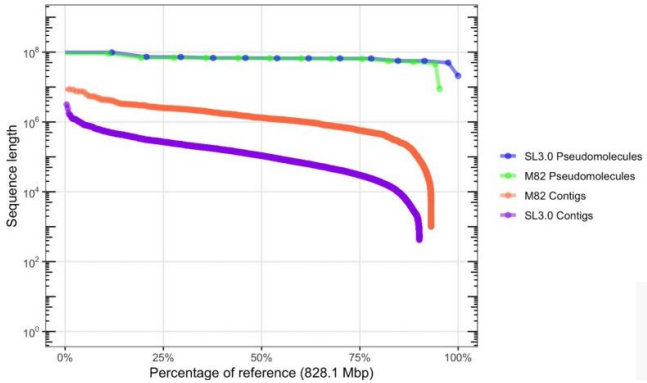
Genome assembly quality control

QV	Mash containment is used to estimate the base error probability. Essentially, uses k-mers found only in the assembly (and not the reads) as an error.
Completeness	Fraction of reliable read k-mers that are also found in the genome assembly

$$\mathrm{QV} = -10 \log_{10} E$$

Genome assembly quality control

Nchart + associated stats



Merqury

Genome Biology

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JOURNAL ARTICLE

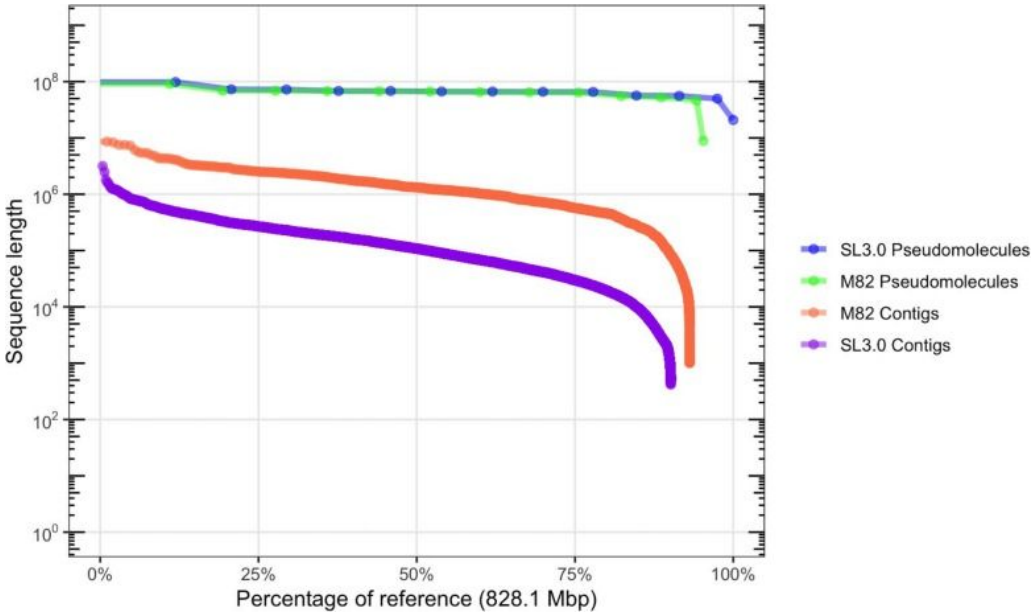
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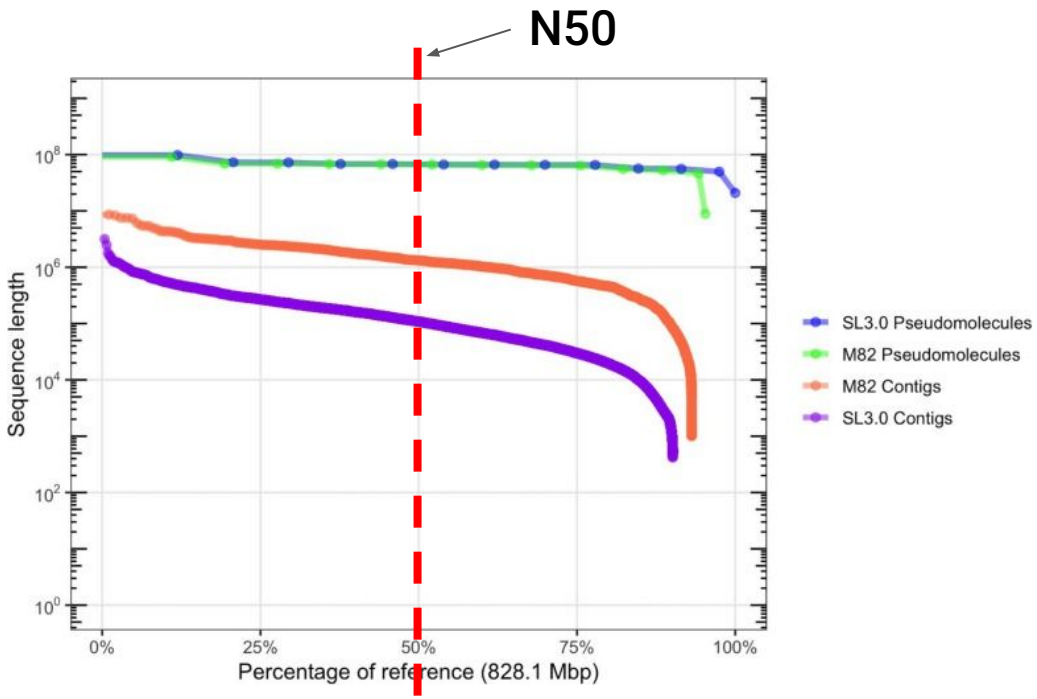
N50 and L50 measure genome assembly contiguity



N50: Length of the contig where 50% of the total assembly is contained in that contig and longer.

In general, higher N50

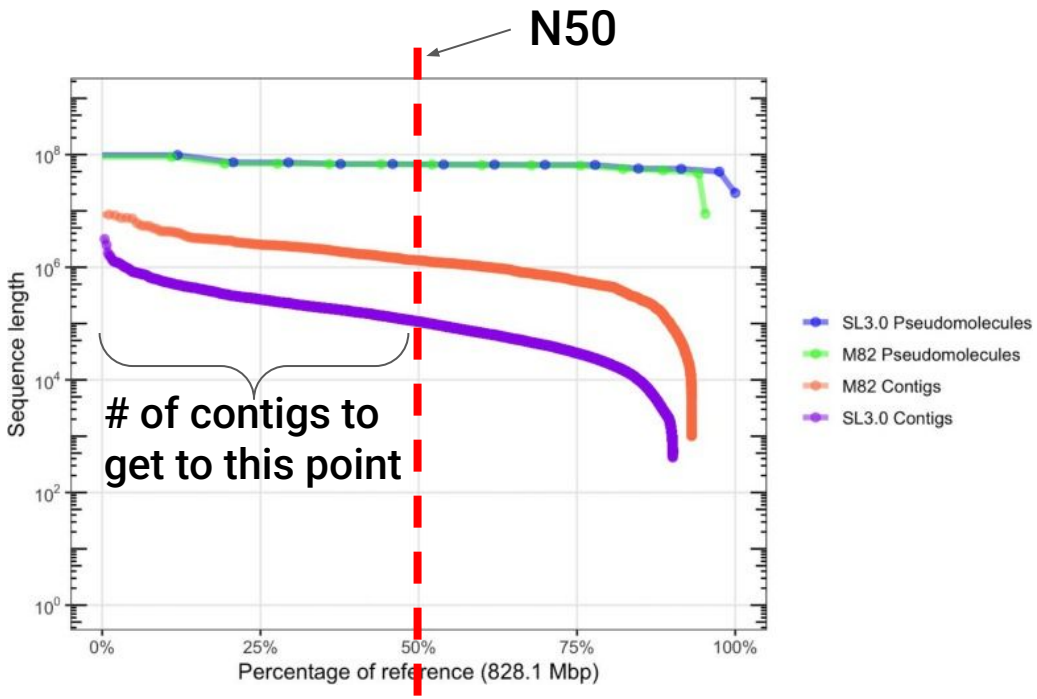
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N50 and L50 measure genome assembly contiguity



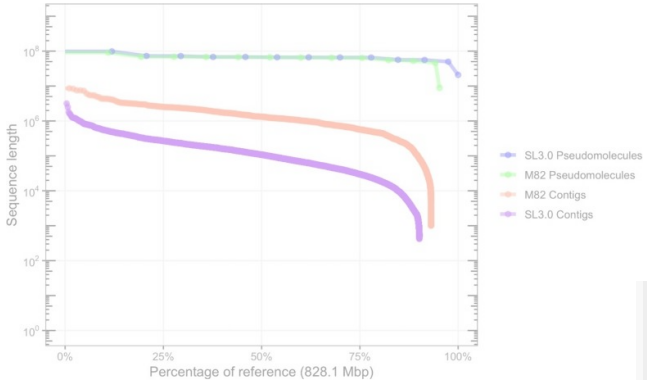
N50: Length of the contig where 50% of the total assembly is contained in that contig and longer.

L50: Minimum number of the longest contigs needed to reach 50% of the total assembly.

In general, higher N50 and lower L50 are desirable

Genome assembly quality control

Nchart + associated stats



Merqury

Genome Biology

Genome Biol. 2020 Sep 14;21:245. doi: 10.1186/s13059-020-02134-9

Merqury: reference-free quality, completeness, and phasing assessment for genome assemblies

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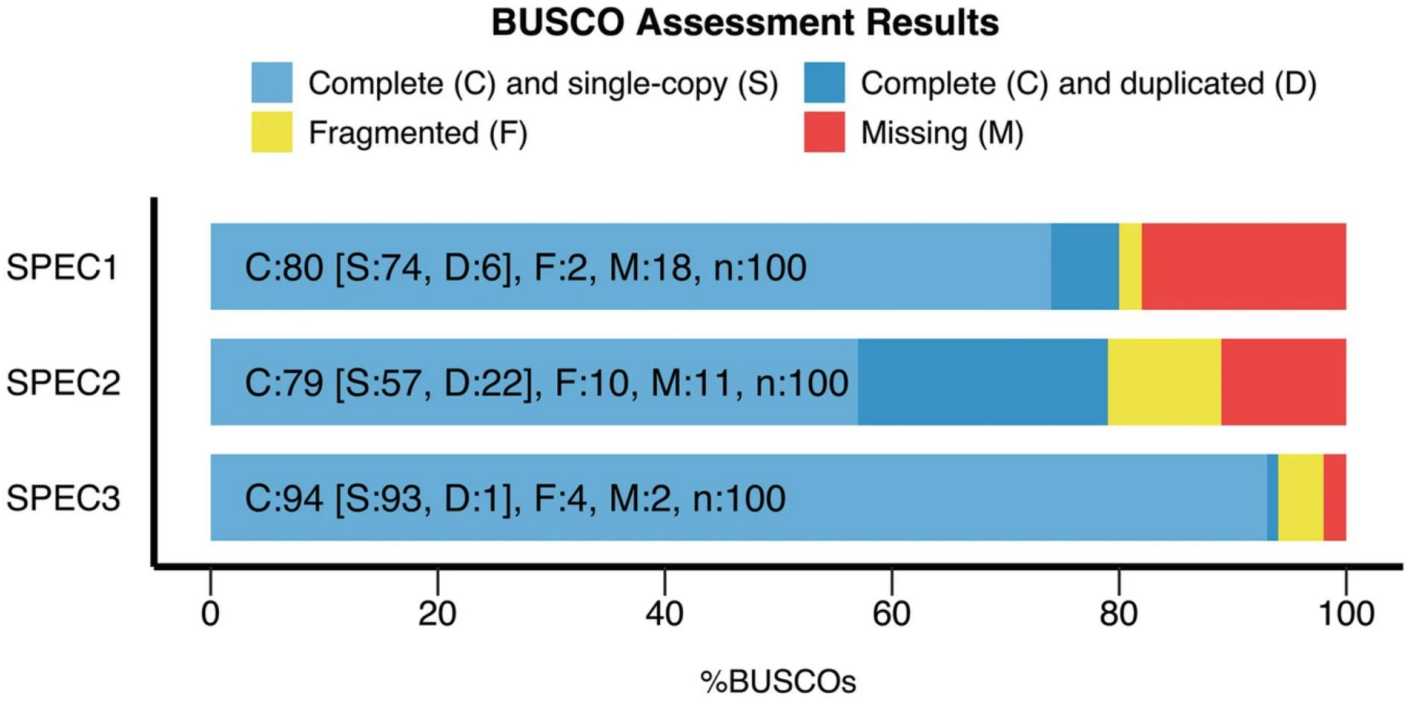
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BUSCOs: Benchmarking universal single copy orthologs

Fig. 3

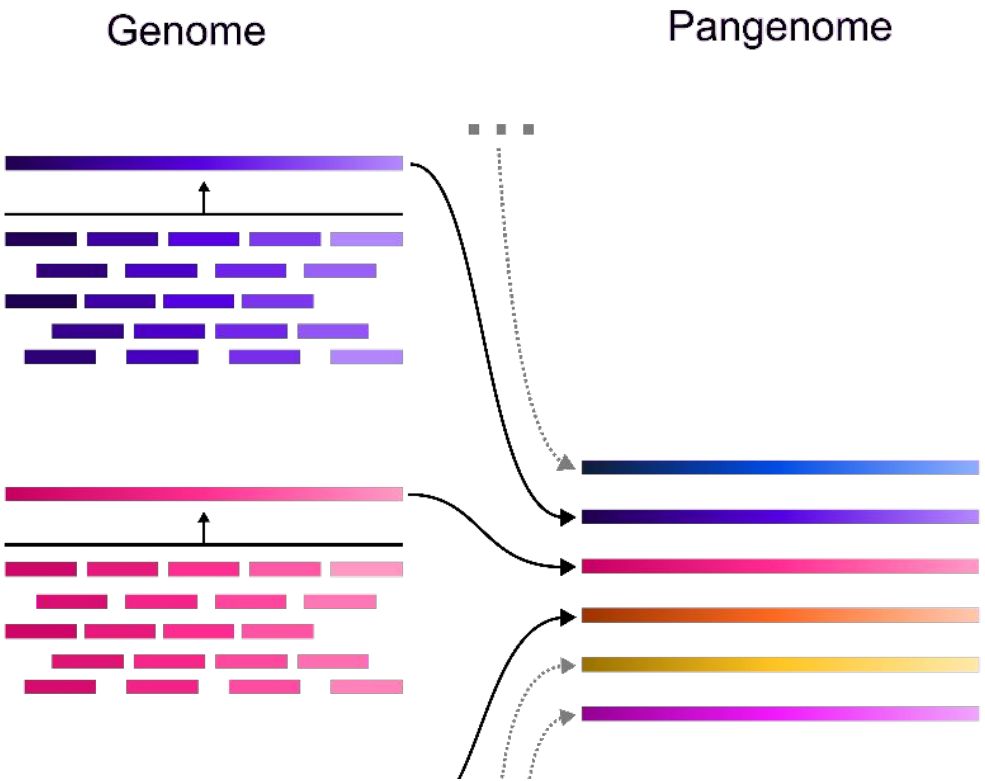
From: **BUSCO: Assessing Genome Assembly and Annotation Completeness**



Outline: Examples of how these tools have been used and combined to answer interesting questions

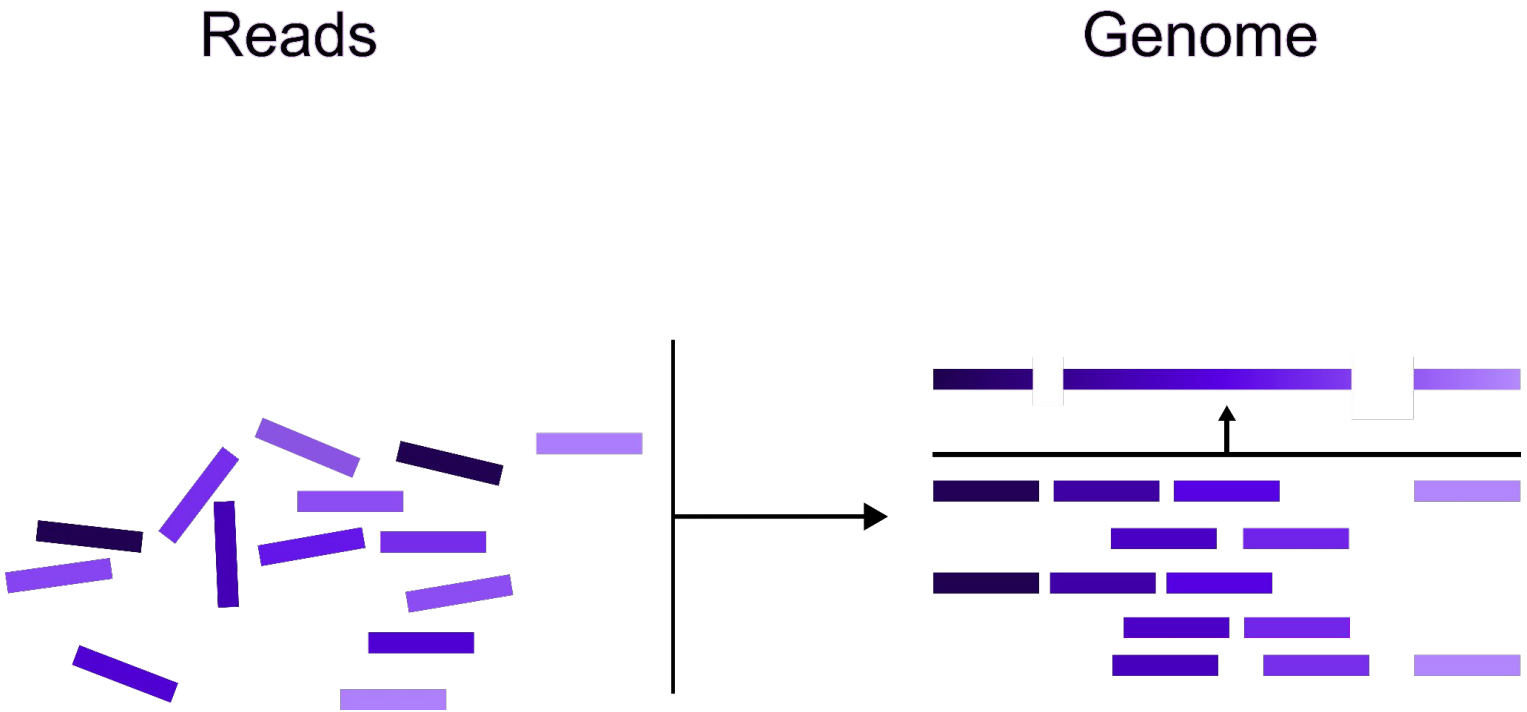
Genome QC

Genome scaffolding

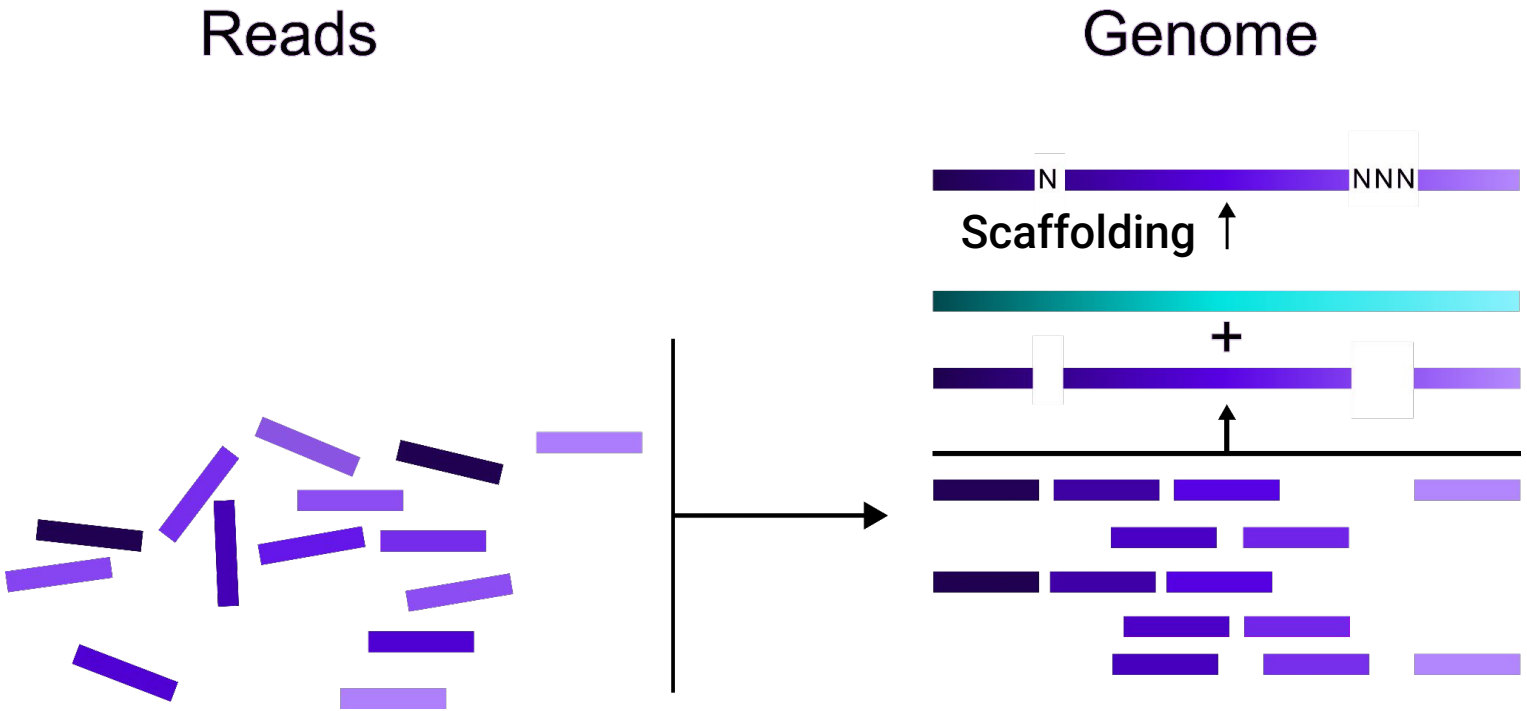


Examples of pangenomes in the wild

Genome scaffolding



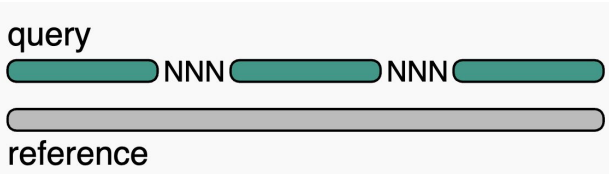
Genome scaffolding



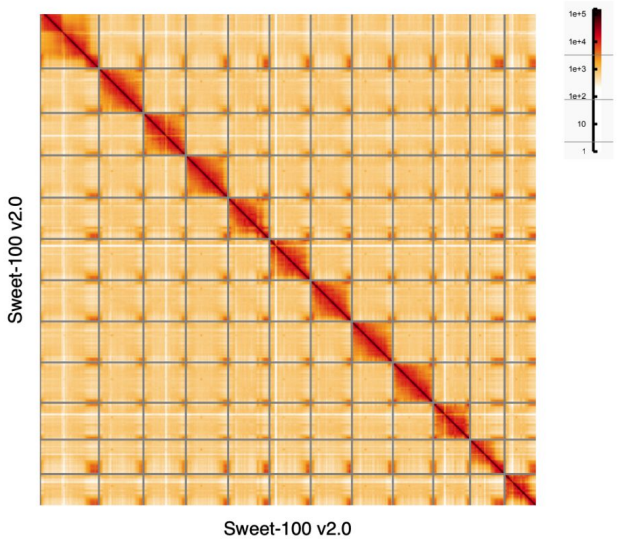
Genome scaffolding

Reference guided

Reference guided

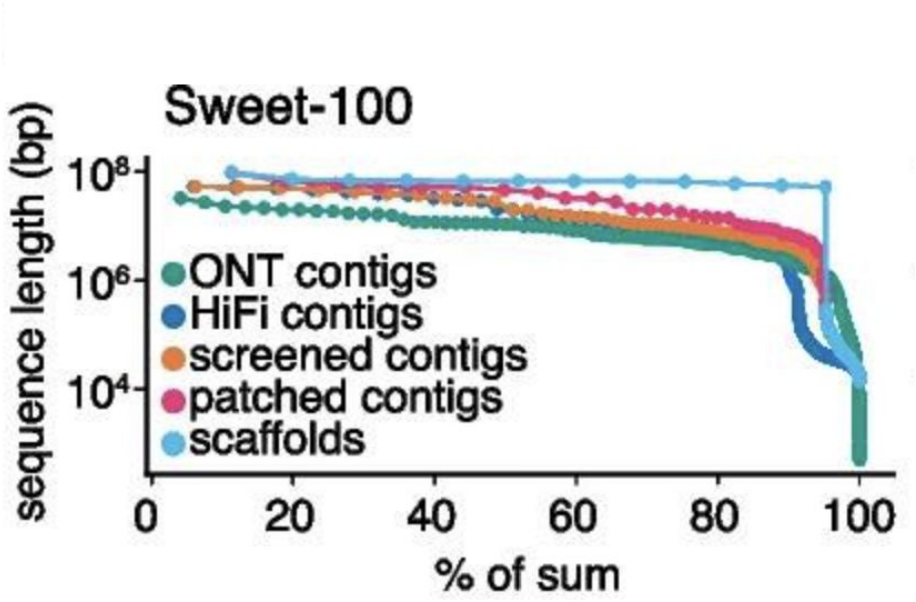


Non reference guided



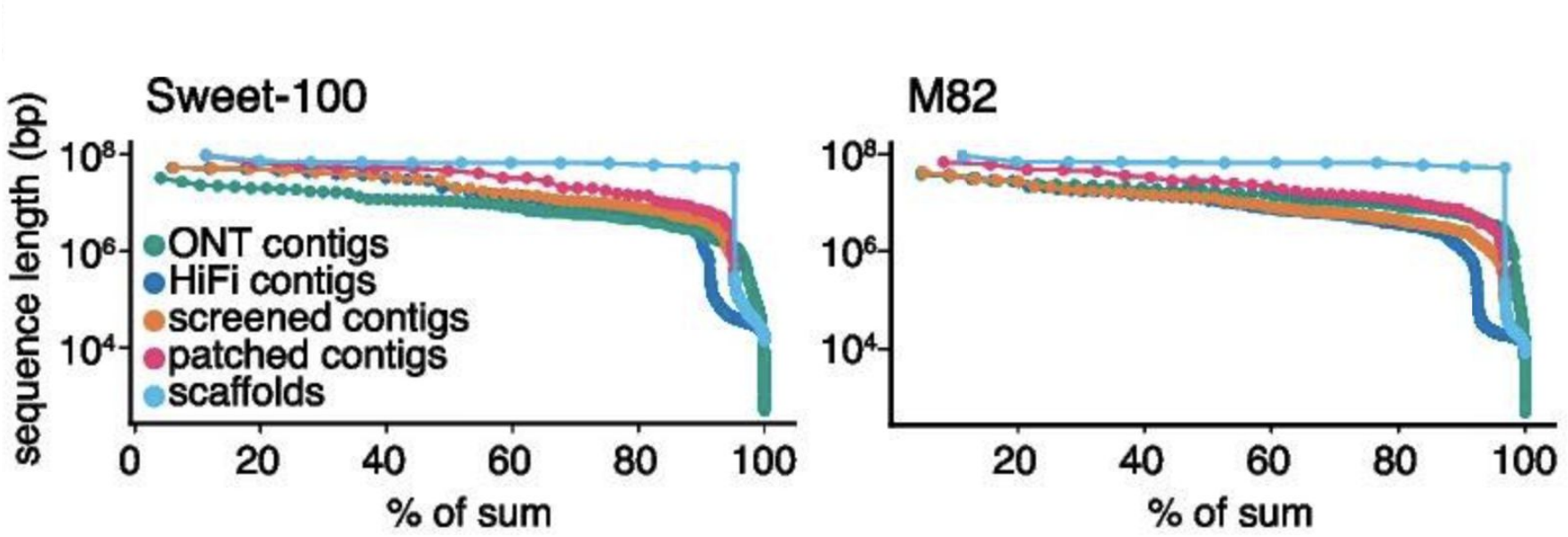
Alonge et al. 2022

Genome scaffolding



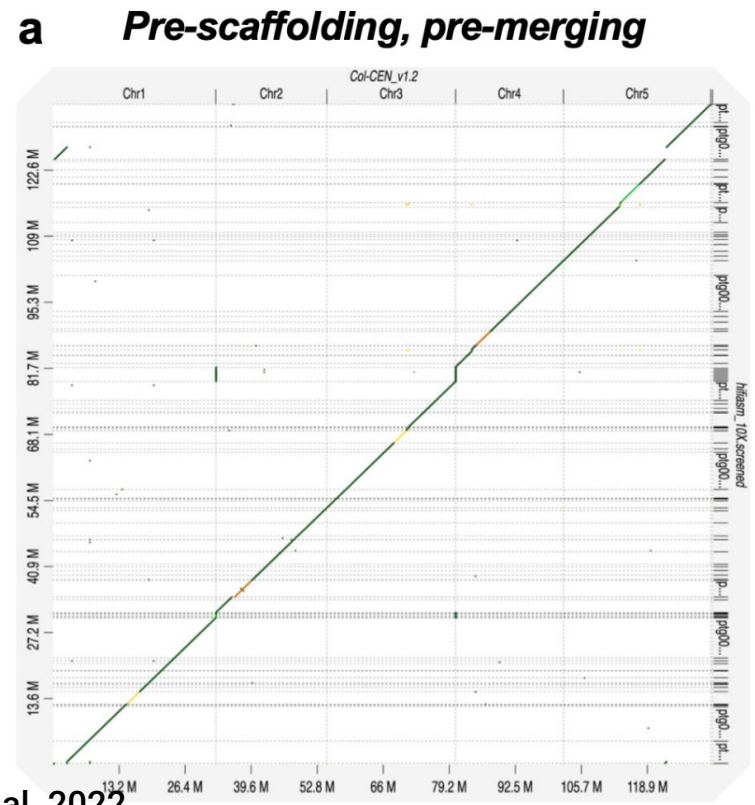
Alonge et al. 2022

Genome scaffolding



Alonge et al. 2022

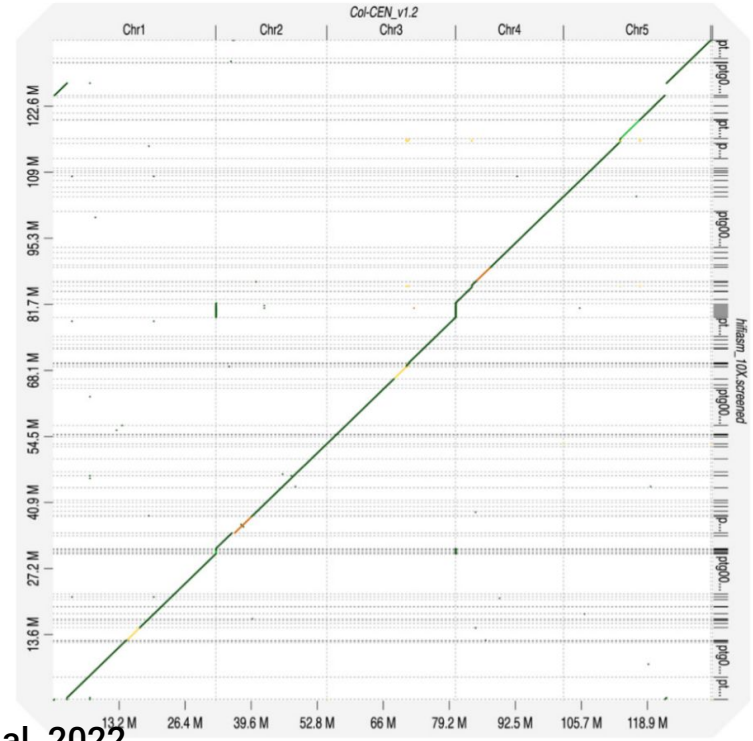
Genome scaffolding



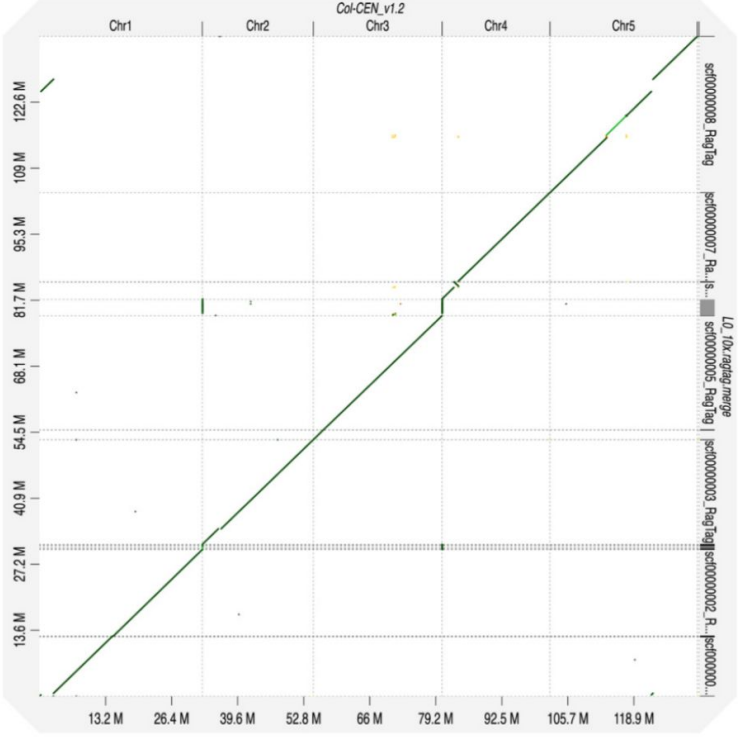
Alonge et al. 2022

Genome scaffolding

a ***Pre-scaffolding, pre-merging***



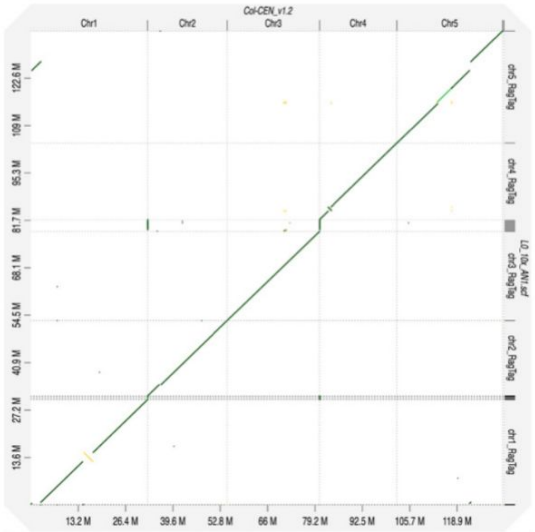
b ***Post-scaffolding, post-merging***



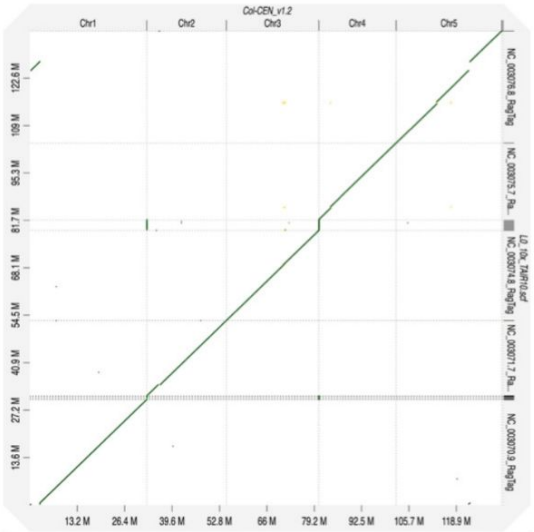
Alonge et al. 2022

Genome scaffolding

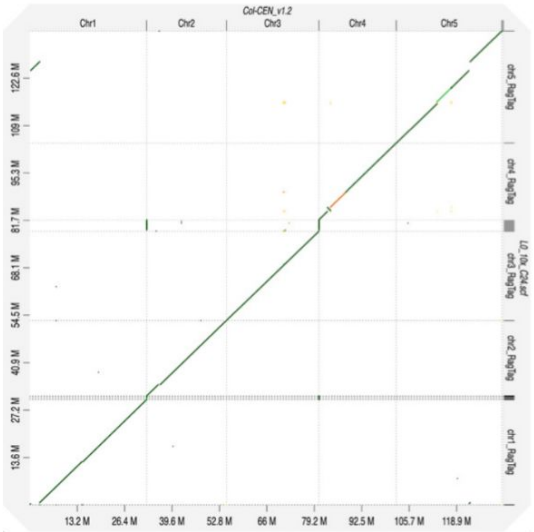
c ***Scaffolding with An-1***



d ***Scaffolding with TAIR10***



e ***Scaffolding with C24***



Alonge et al. 2022

Genome scaffolding

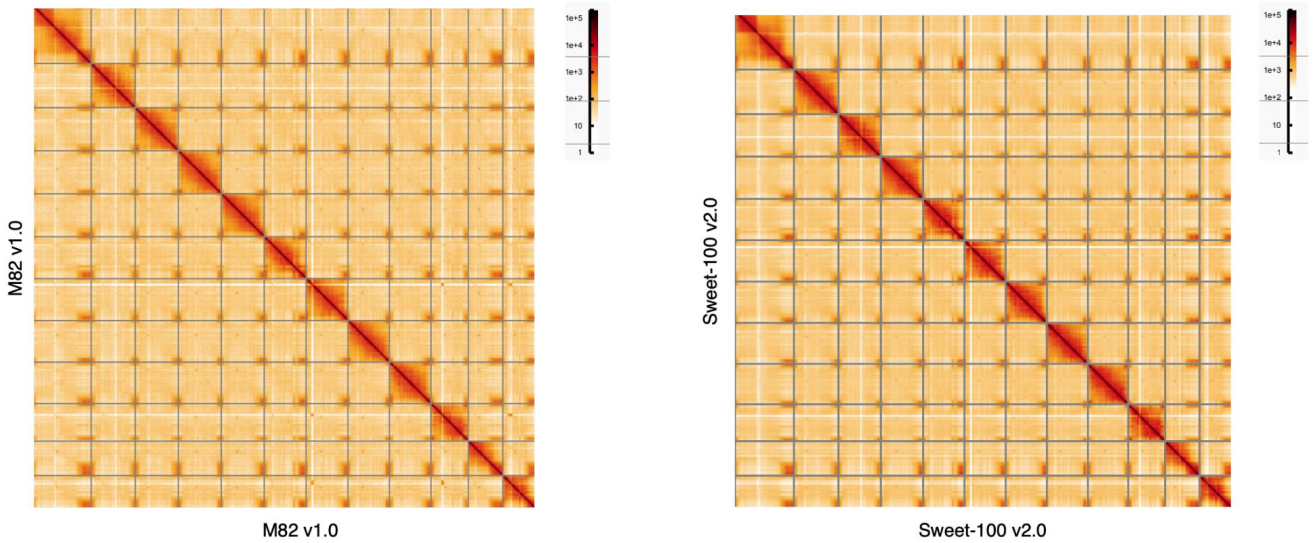


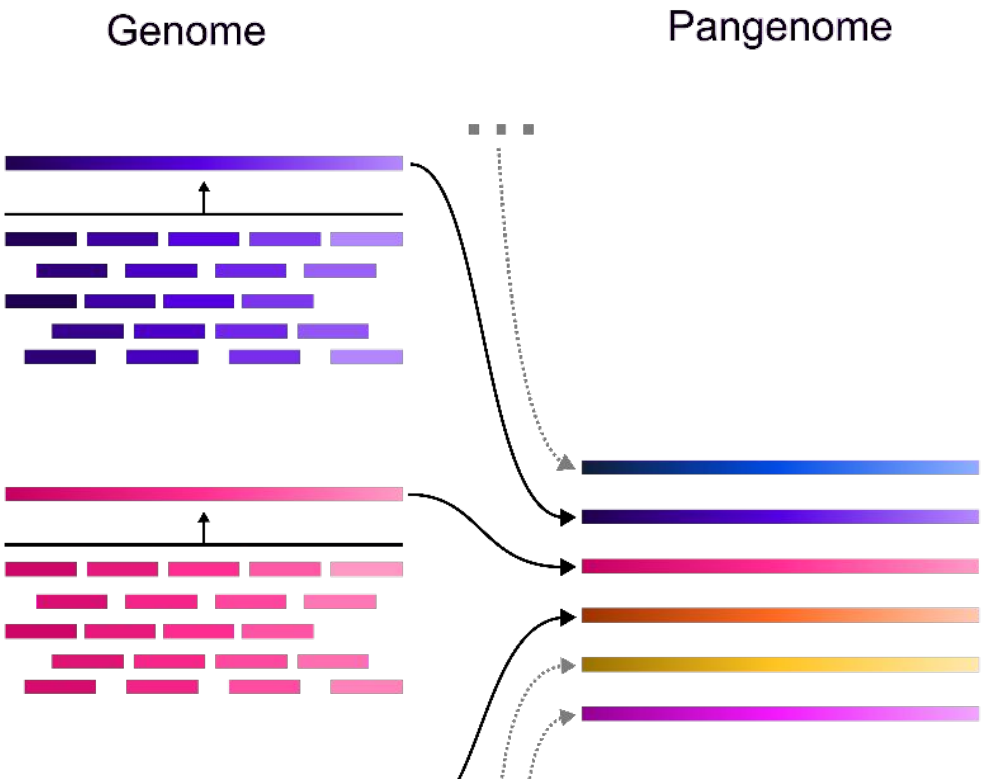
Fig. S12 | Hi-C heatmaps confirm reference assembly structural accuracy. Hi-C heatmaps for the M82 v1.0 and the Sweet-100 v2.0 reference assemblies. The 12 chromosomes are sorted from largest (top left) to smallest (bottom right).

Alonge et al. 2022

Outline: Examples of how these tools have been used and combined to answer interesting questions

Genome QC

Genome scaffolding



Examples of pangenomes in the wild

Case studies

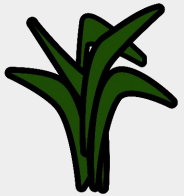
1. Multiple jay species

- Full genome assemblies
- Linked population size to SVs



2. Rice GWAS

- Available rice assemblies
- Short read sequencing from >400 accessions
- GWAS to identify plant height loci



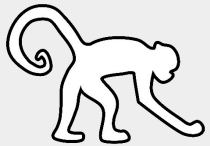
3. Tomato pangenome

- SVs from long reads
- Mix of assemblies (14) and raw reads from a single species
- Identified tomato flavor important loci



4. *Pan* pangenome

- Multi species pangenome
- Constructed several near T2T genomes
- Insights into the evolutionary history and adaptation of humans



Case study 1: Scrub jay pangenome

RESEARCH ARTICLE SUMMARY

GENOMICS

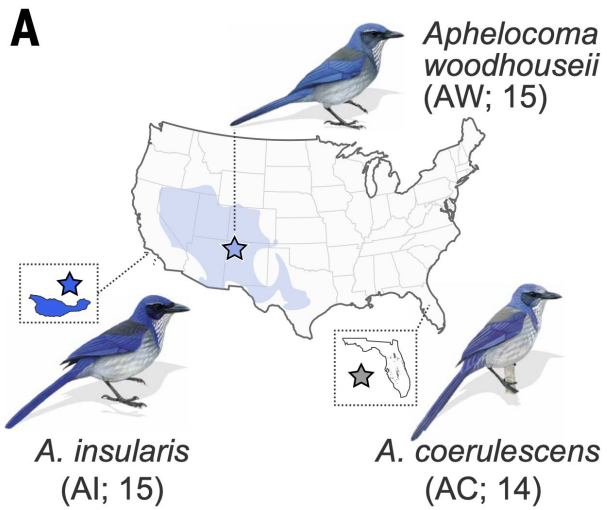
Multispecies pangenomes reveal a pervasive influence of population size on structural variation

Scott V. Edwards*, Bohao Fang, Danielle Khost, George E. Kolyfetis, Rebecca G. Cheek, Devon A. DeRaad, Nancy Chen, John W. Fitzpatrick, John E. McCormack, W. Chris Funk, Cameron K. Ghalambor, Erik Garrison, Andrea Guarracino, Heng Li, Timothy B. Sackton

Full article and list of author affiliations: https://doi.org/10.1126/science.adw1931

Edwards et al. 2025

Case study 1: Scrub jay pangenome

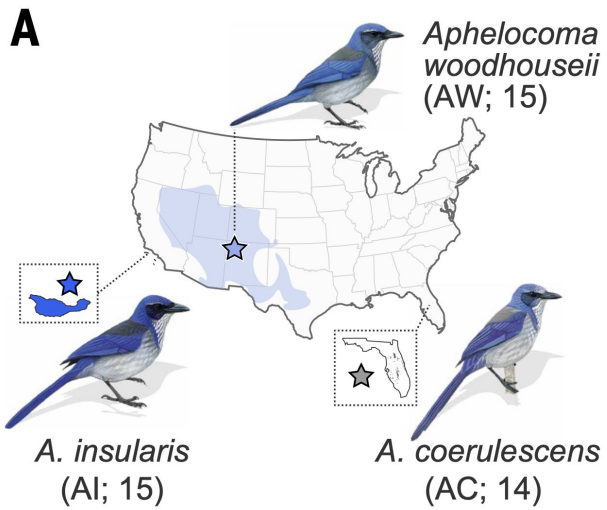


Data:

- Pacbio sequencing from three populations of scrub jay, with variable population sizes.

Edwards et al. 2025

Case study 1: Scrub jay pangenome



Data:

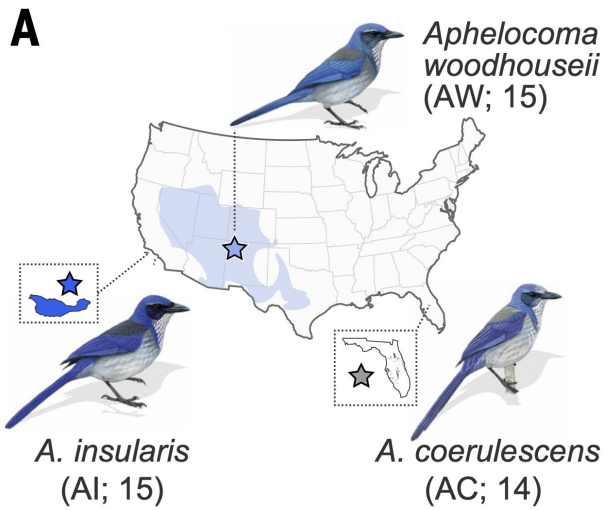
- Pacbio sequencing from three populations of scrub jay, with variable population sizes.

Methods:

- Hifiasm for the genome assembly
- BUSCO completeness for QC
- PGGB and minigraph for graph construction
- ModDotPlot for repeat visualization

Edwards et al. 2025

Case study 1: Scrub jay pangenome



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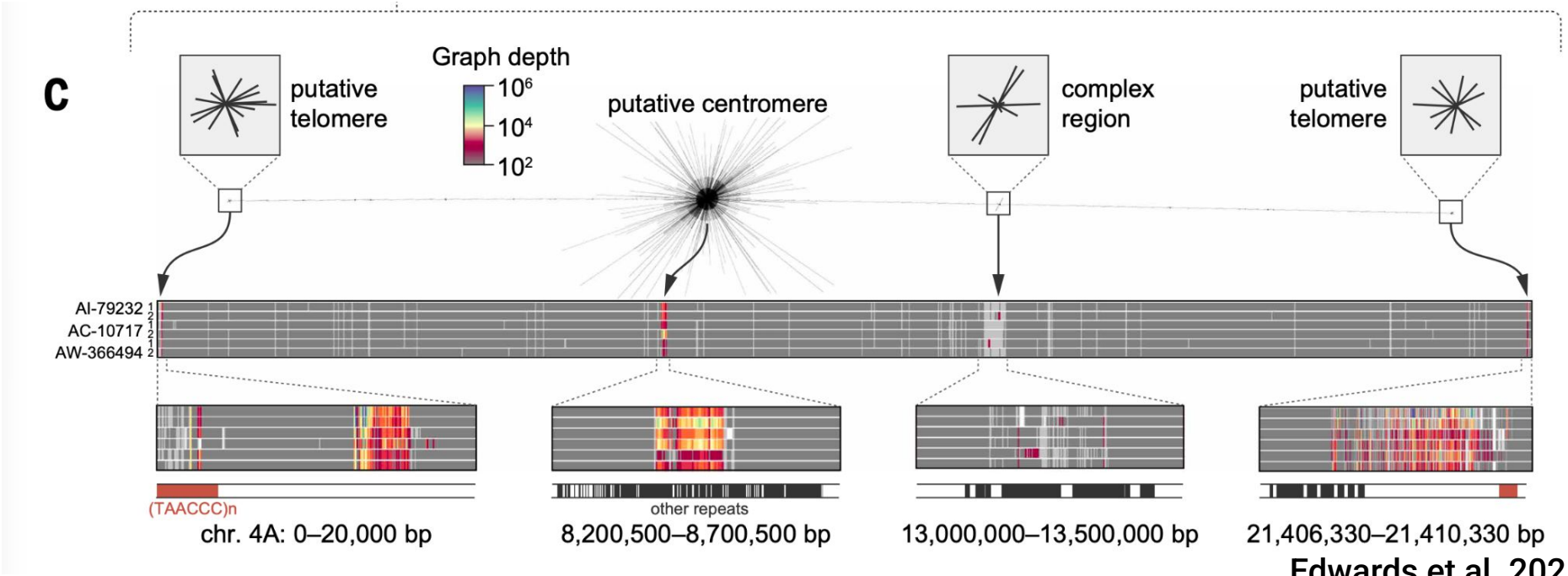
- Hifiasm for the genome assembly
- BUSCO completeness for QC
- PGGB and minigraph for graph construction
- ModDotPlot for repeat visualization

Main results:

- Found rapid evolution of genome structure
- Structural changes driven by satellite DNA shifts
- Gene copy number variation is inversely proportional to population size Edwards et al. 2025

Case study 1: Scrub jay pangenome

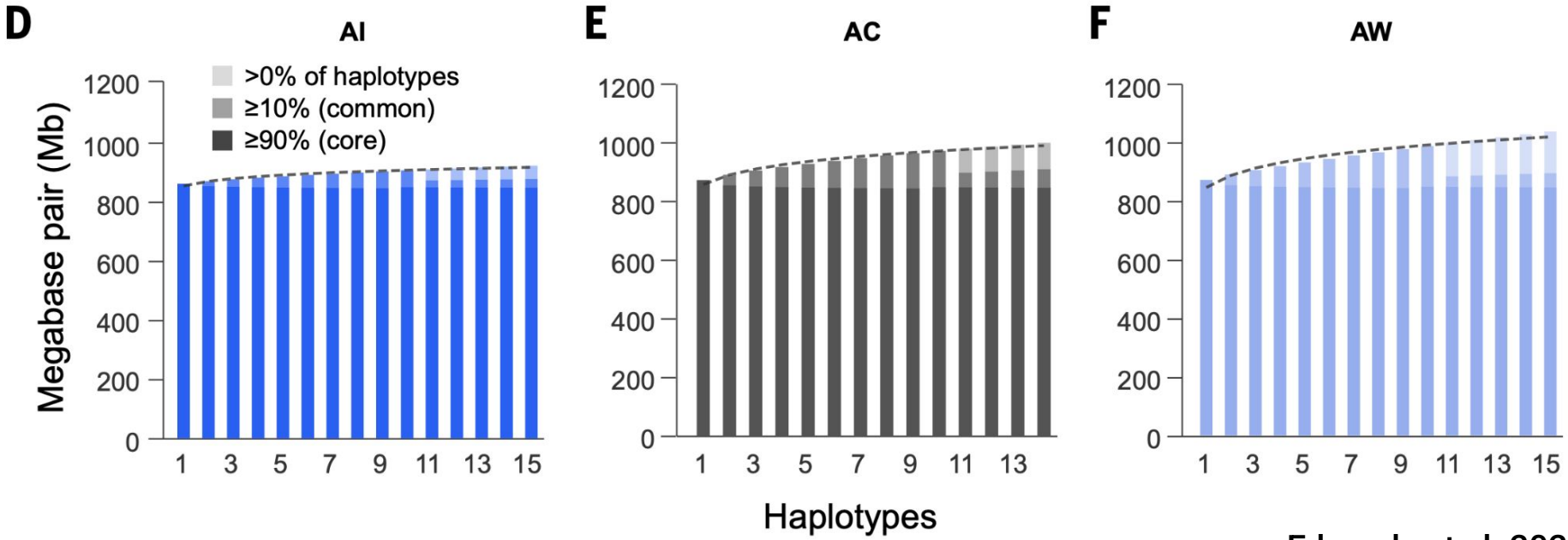
PGGB graph (annotated)



Edwards et al. 2025

Case study 1: Scrub jay pangenome

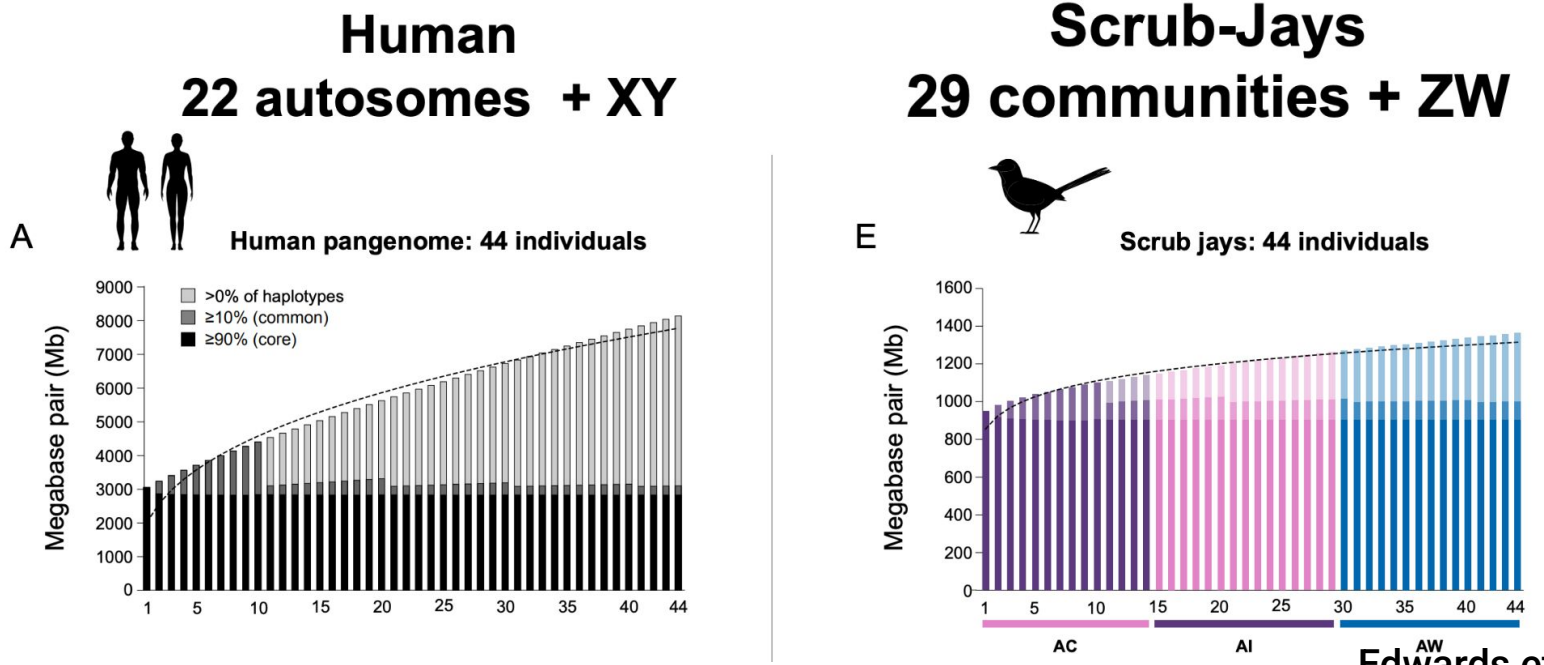
Panacus



Edwards et al. 2025

Case study 1: Scrub jay pangenome

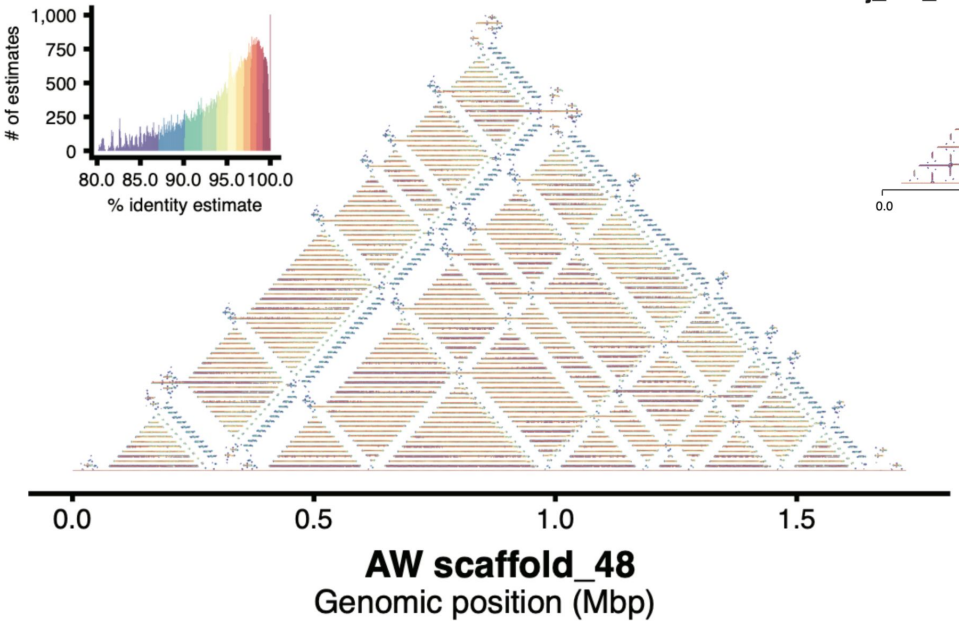
Panacus

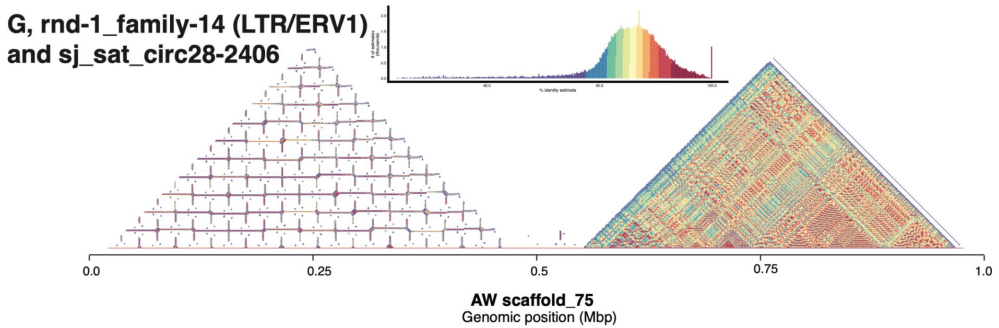


Edwards et al. 2025

Case study 1: Scrub jay pangenome

ModDotPlot for repeat exploration

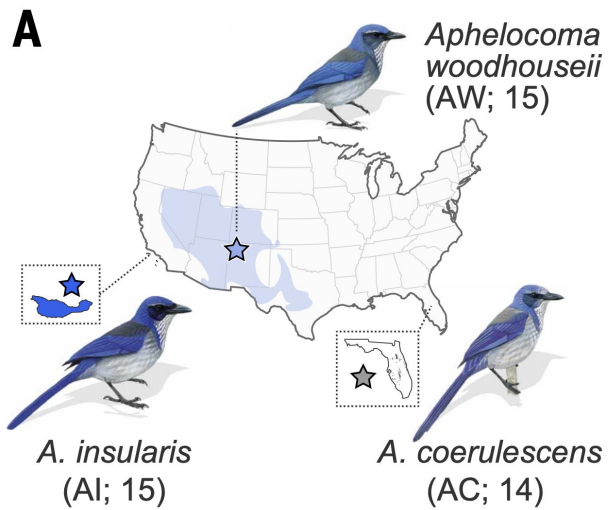


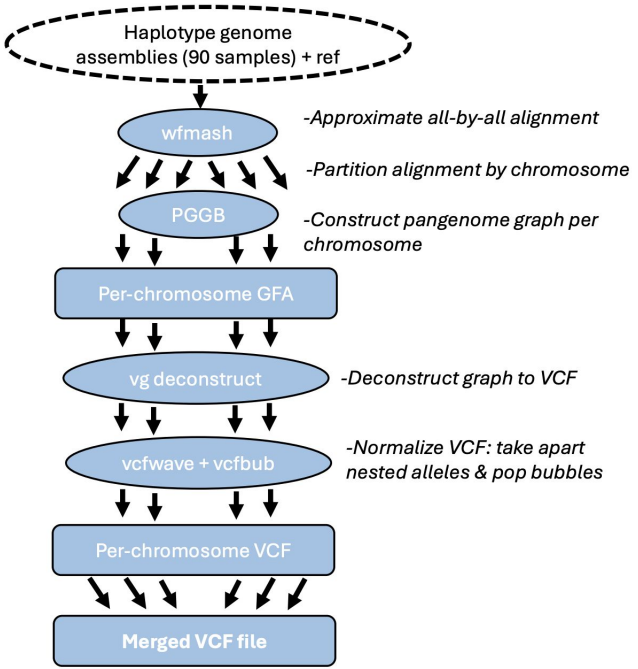


Edwards et al. 2025

Case study 1: Scrub jay pangenome

Variant calling pipeline

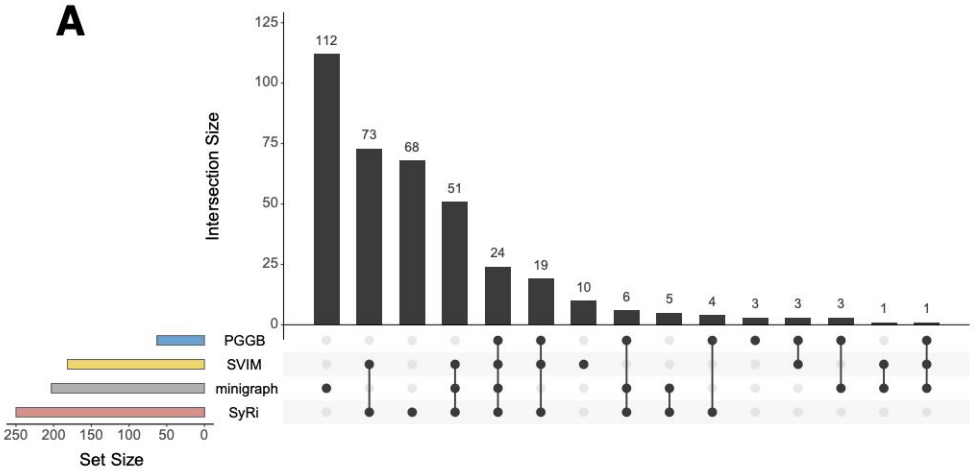


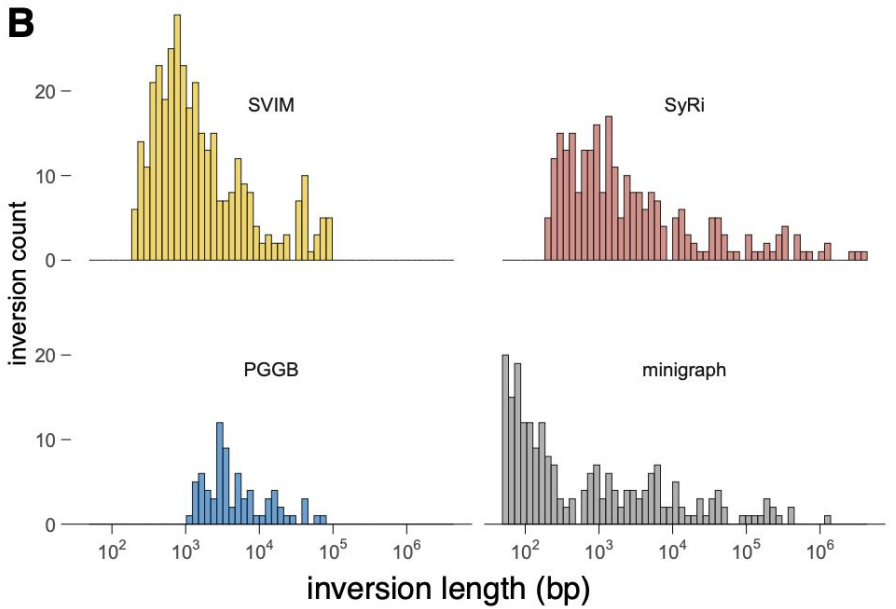


Edwards et al. 2025

Case study 1: Scrub jay pangenome

Variant calling comparison





For SV analysis, required at least 3 methods to agree

Edwards et al. 2025

Case studies

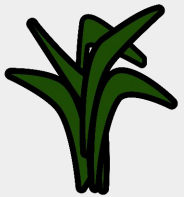
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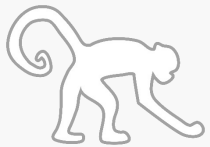
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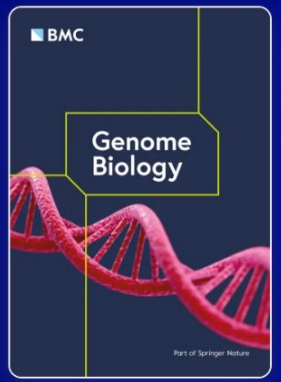
Case study 2: Rice pangenome GWAS

Home > Genome Biology > Article

A pangenome analysis pipeline provides insights into functional gene identification in rice

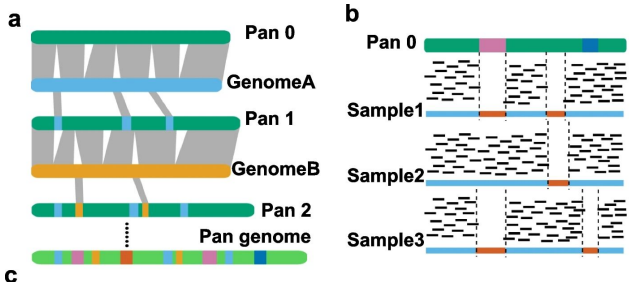
Research | Open access | Published: 26 January 2023

Volume 24, article number 19 (2023) Cite this article



Genome Biology

Case study 2: Rice pangenome GWAS

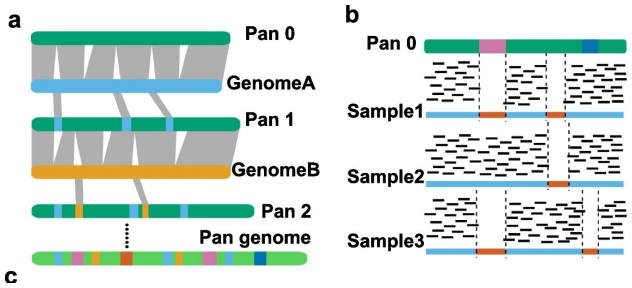


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- 413 rice panel samples (short reads)

Wang et al. 2023

Case study 2: Rice pangenome GWAS



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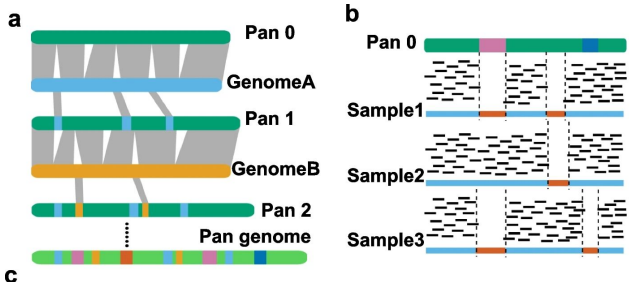
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Wang et al. 2023

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Main results:

- Identified casual SVs for grain weight and plant height via GWAS

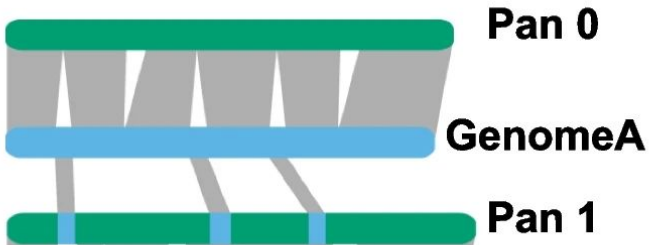
Wang et al. 2023

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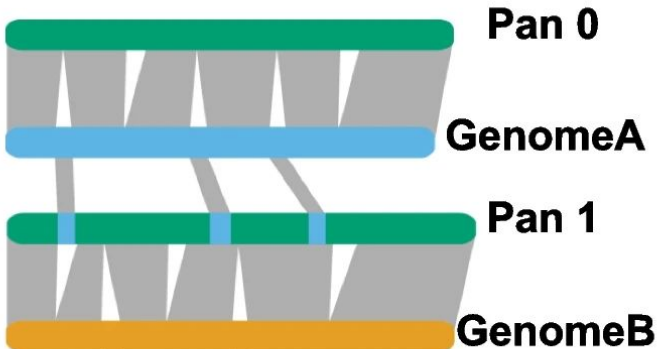
Wang et al. 2023

Case study 2: Rice pangenome GWAS



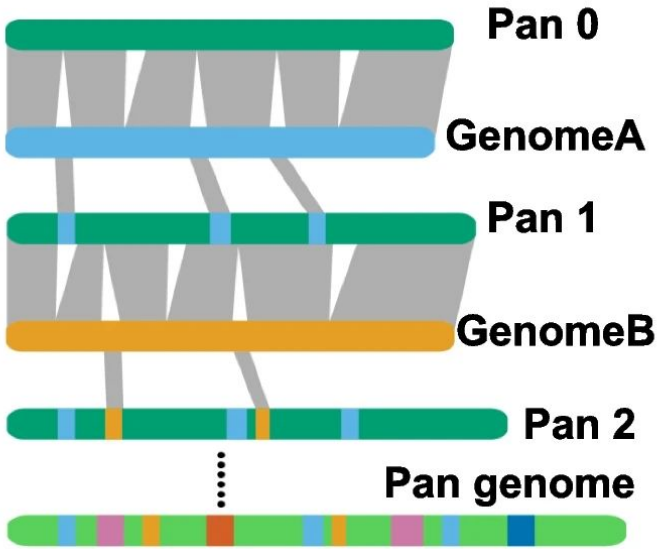
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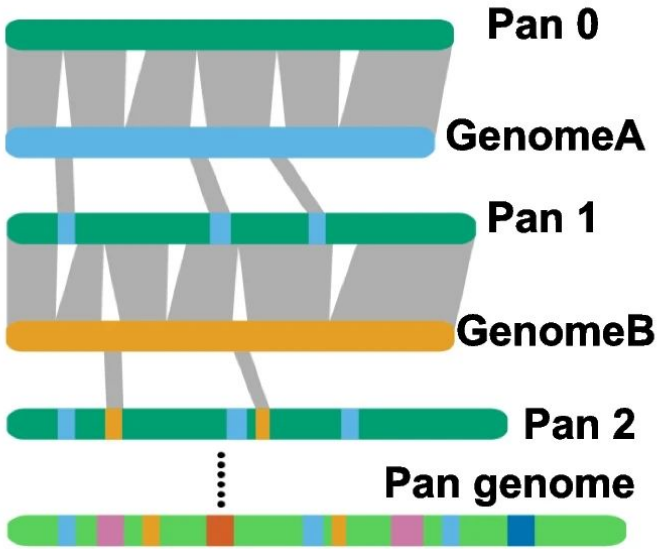
Wang et al. 2023

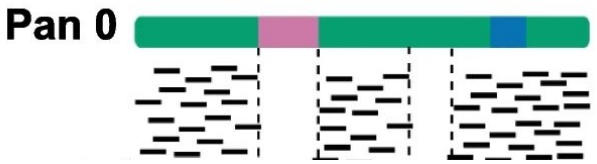
Case study 2: Rice pangenome GWAS



Wang et al. 2023

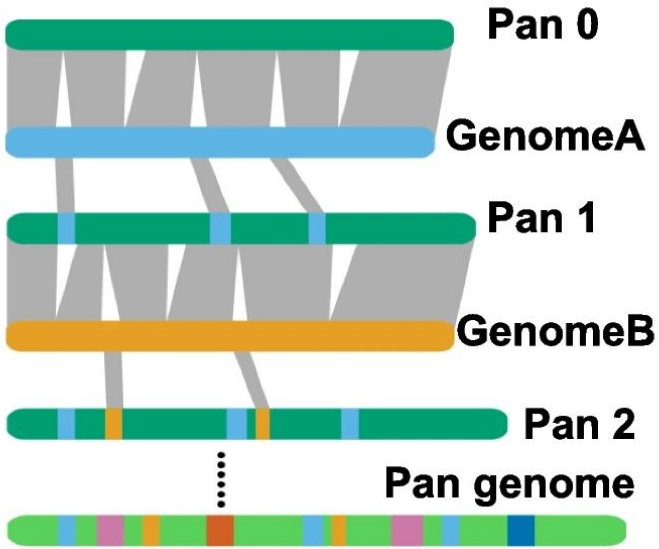
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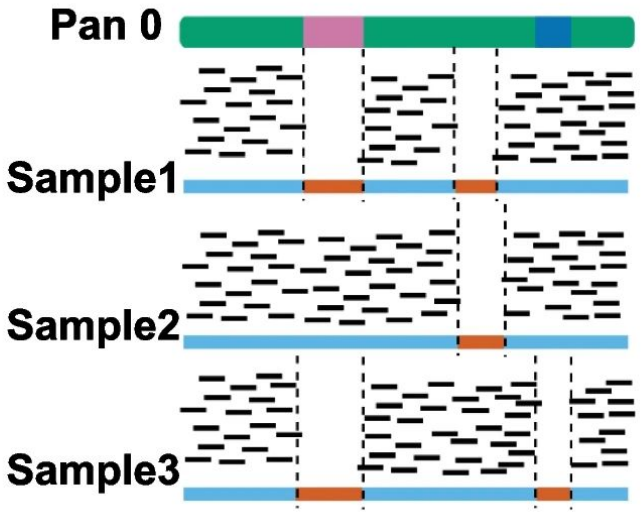




Wang et al. 2023

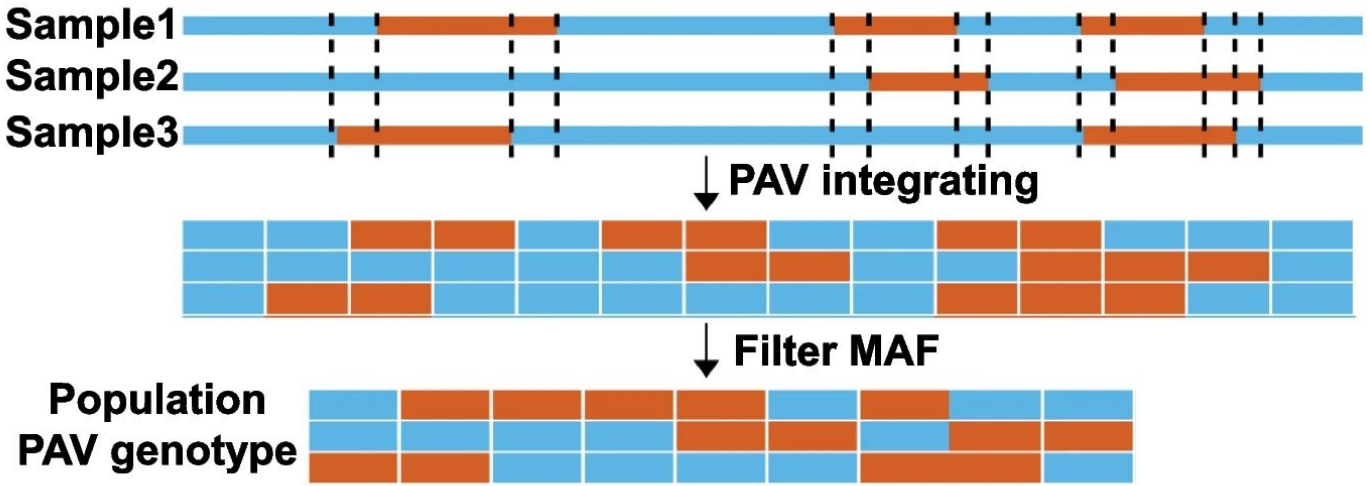
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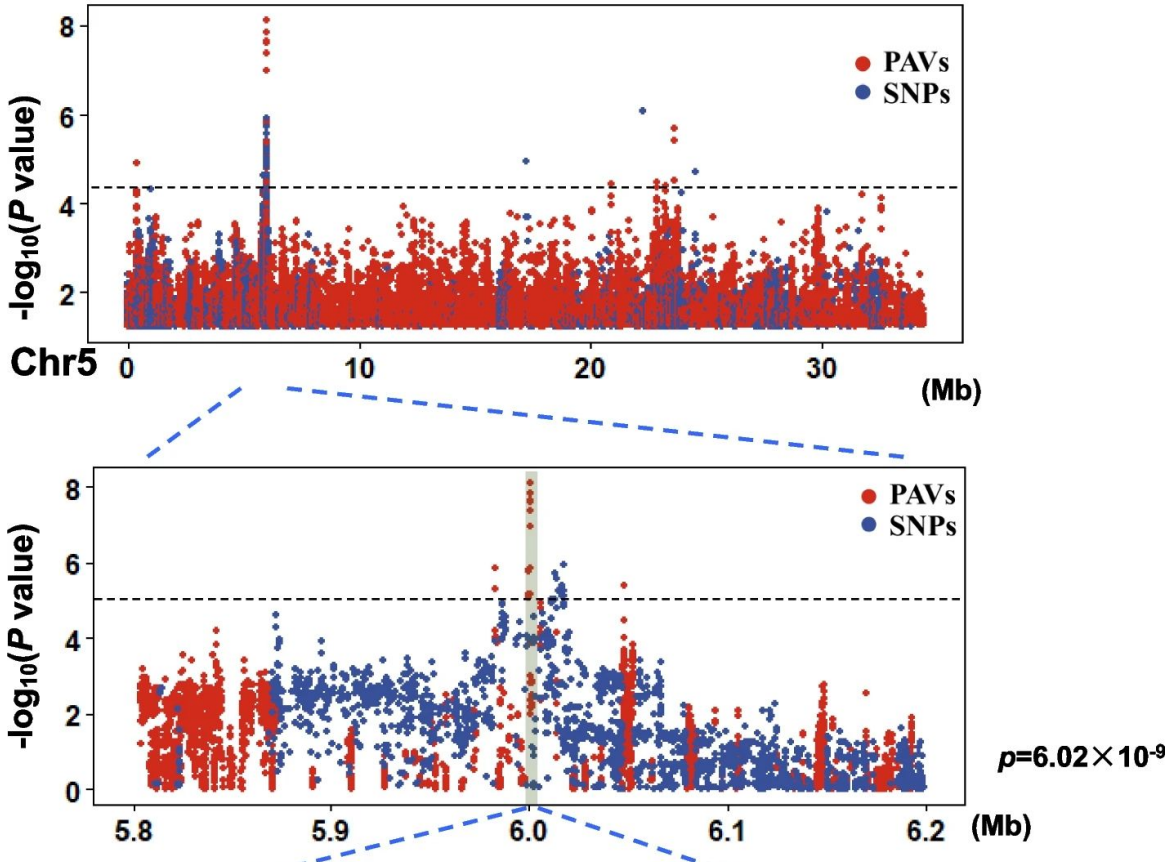
Wang et al. 2023

Case study 2: Rice pangenome GWAS



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Wang et al. 2023

Case studies

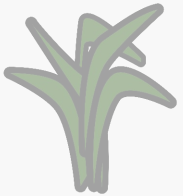
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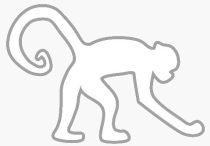
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Case study 3: 100 tomatoes



> Cell. 2020 Jul 9;182(1):145-161.e23. doi: 10.1016/j.cell.2020.05.021. Epub 2020 Jun 17.

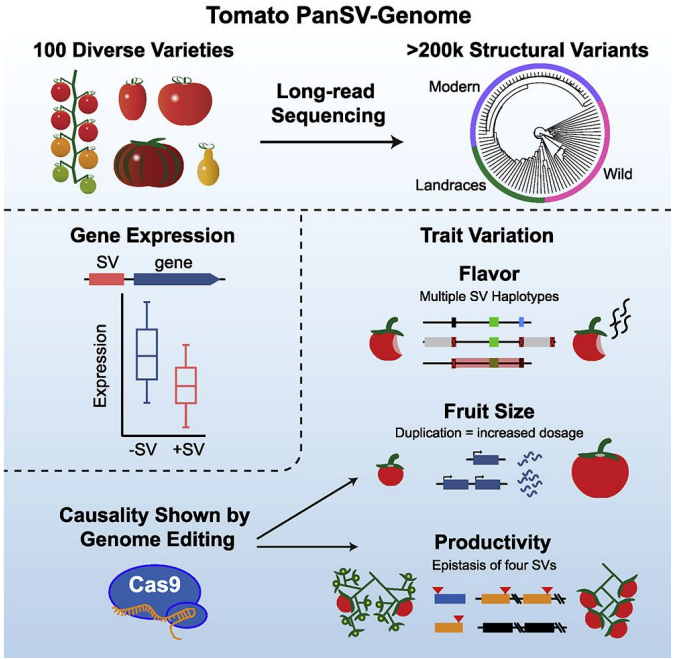
Major Impacts of Widespread Structural Variation on Gene Expression and Crop Improvement in Tomato

Michael Alonge [1], Xingang Wang [2], Matthias Benoit [3], Sebastian Soyk [2], Lara Pereira [4], Lei Zhang [4], Hamsini Suresh [2], Srividya Ramakrishnan [1], Florian Maumus [5], Danielle Ciren [2], Yuval Levy [6], Tom Hai Harel [6], Gili Shalev-Schlosser [6], Ziva Amsellem [6], Hamid Razifard [7], Ana L Caicedo [7], Denise M Tieman [8], Harry Klee [8], Melanie Kirsche [1], Sergey Aganezov [1], T Rhyker Ranallo-Benavidez [9], Zachary H Lemmon [2], Jennifer Kim [3], Gina Robitaille [3], Melissa Kramer [2], Sara Goodwin [2], W Richard McCombie [10], Samuel Hutton [11], Joyce Van Eck [12], Jesse Gillis [2], Yuval Eshed [6], Fritz J Sedlazeck [13], Esther van der Knaap [14], Michael C Schatz [15], Zachary B Lippman [16]

Affiliations + expand

PMID: 32553272 PMCID: PMC7354227 DOI: 10.1016/j.cell.2020.05.021

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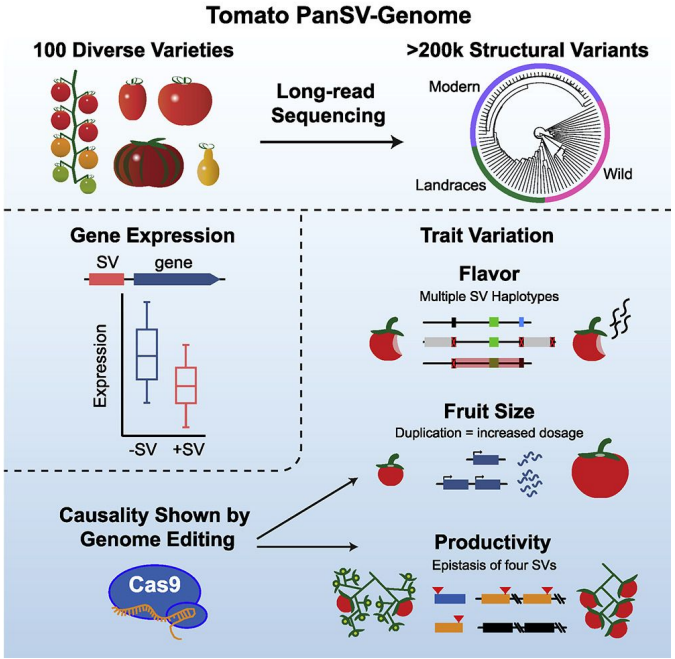


Data:

- 100 tomatoes with ONT reads and illumina reads

Alonge et al. 2020

Case study 3: 100 tomatoes



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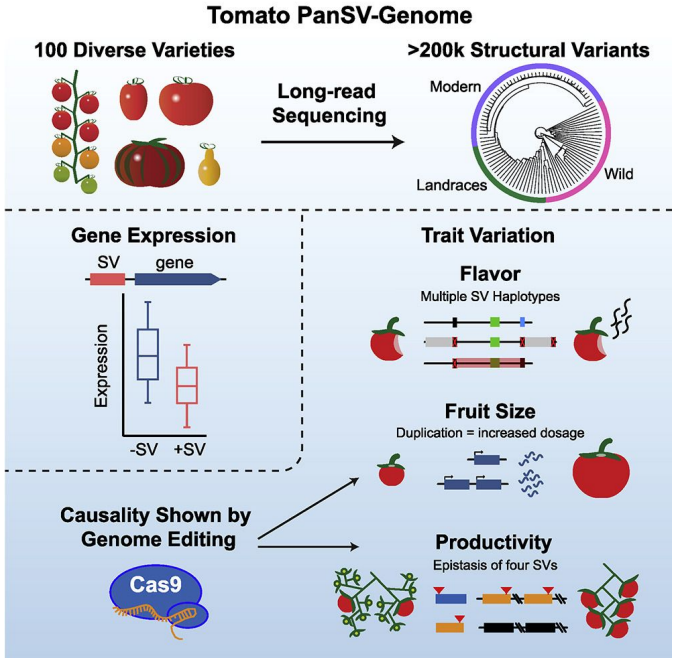
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- Generated 14 assemblies with MaSuRCA
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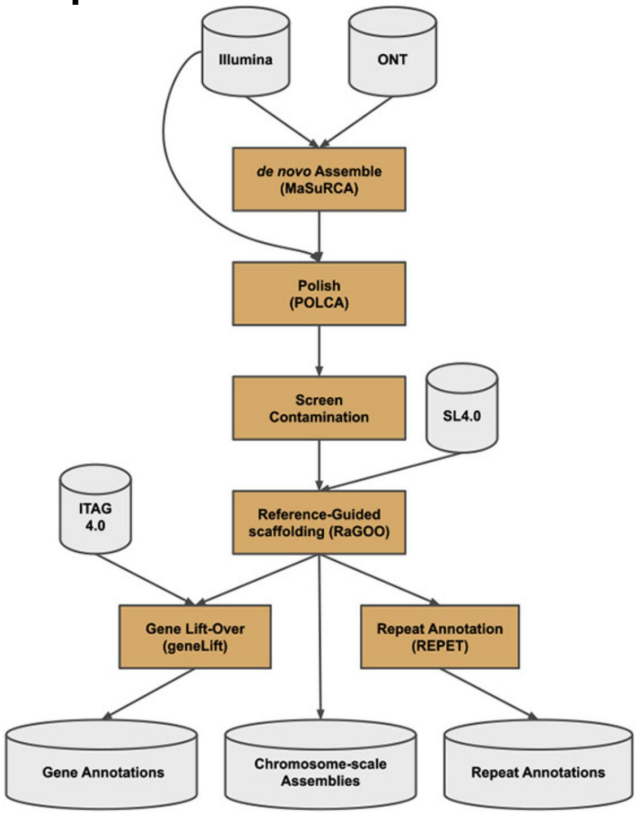
- Identified introgressions in specific tomato groups
- Identified an SV that impacts tomato smoky flavor
- Identified SVs associated with productivity
- Validated with genome editing

Alonge et al. 2020

Case study 3: 100 tomatoes



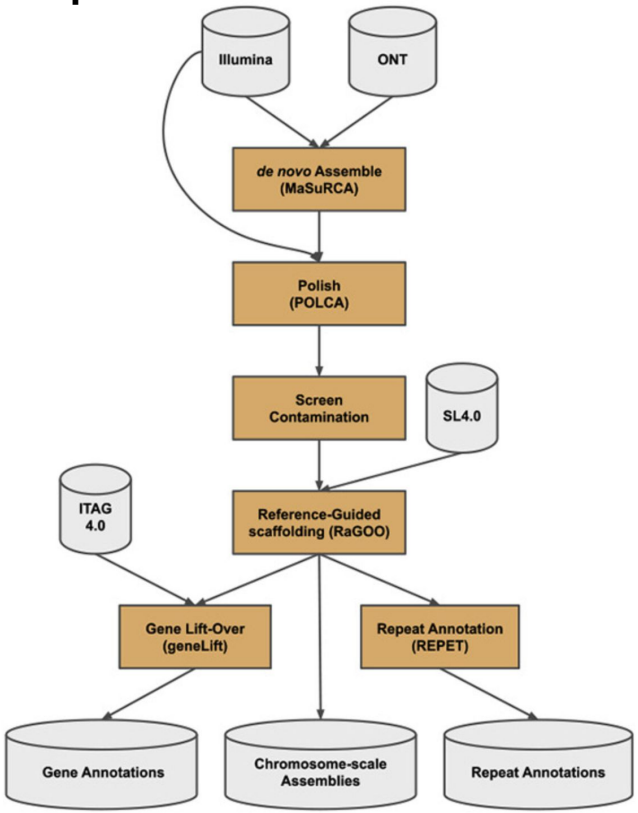
Pipeline



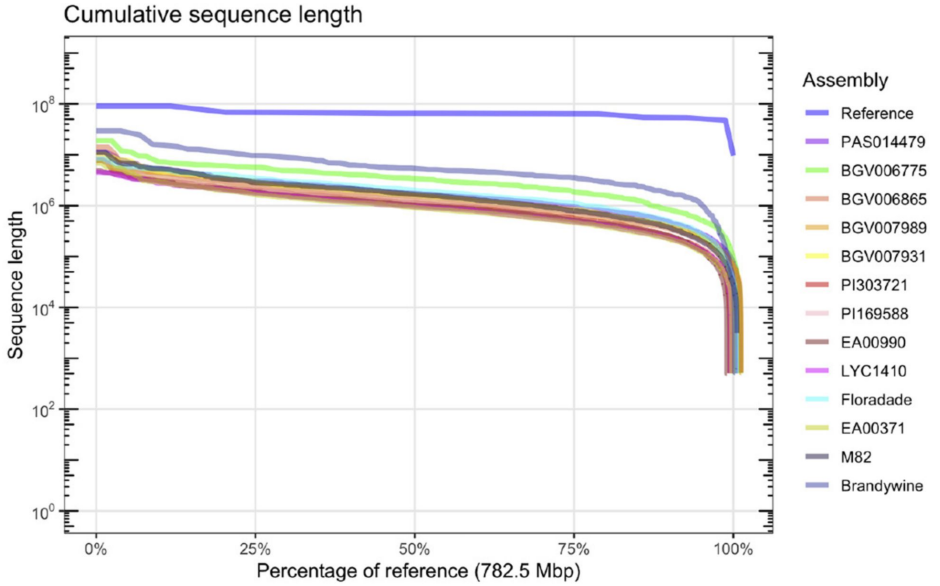
Alonge et al. 2020

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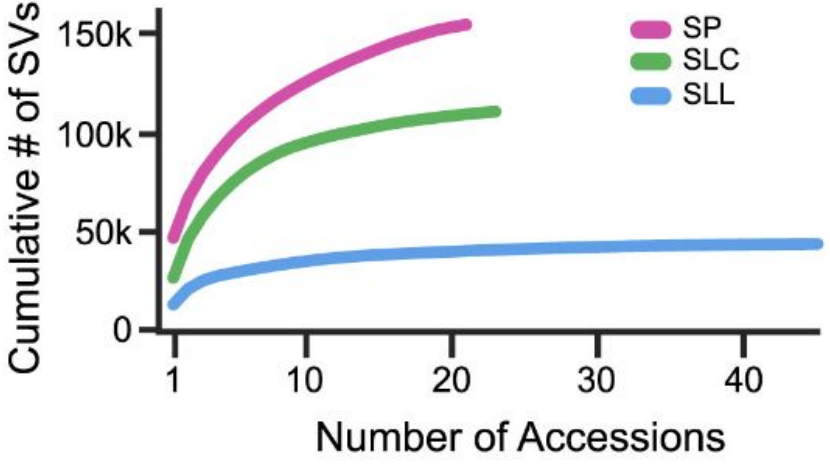
NChart (QC)



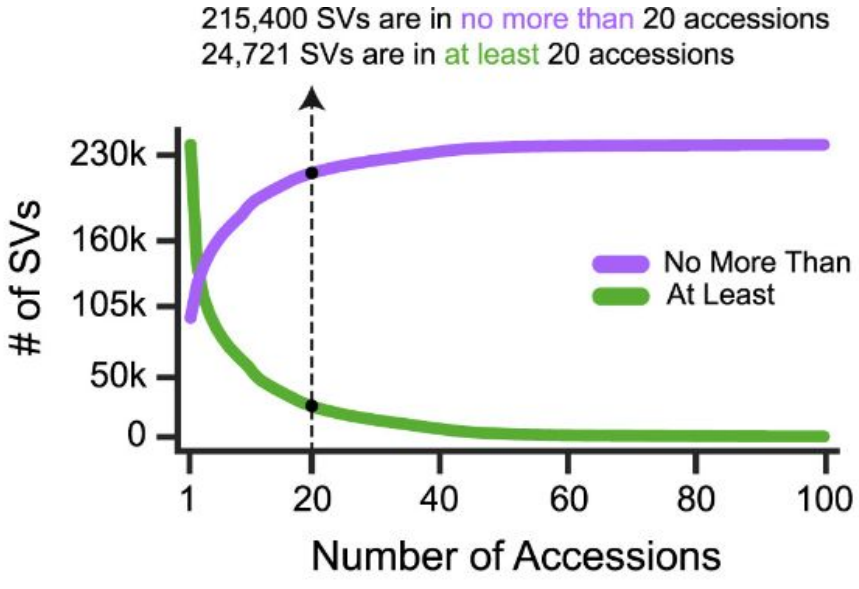
Alonge et al. 2020

Case study 3: 100 tomatoes

Pangenome growth curve



Pangenome core curve

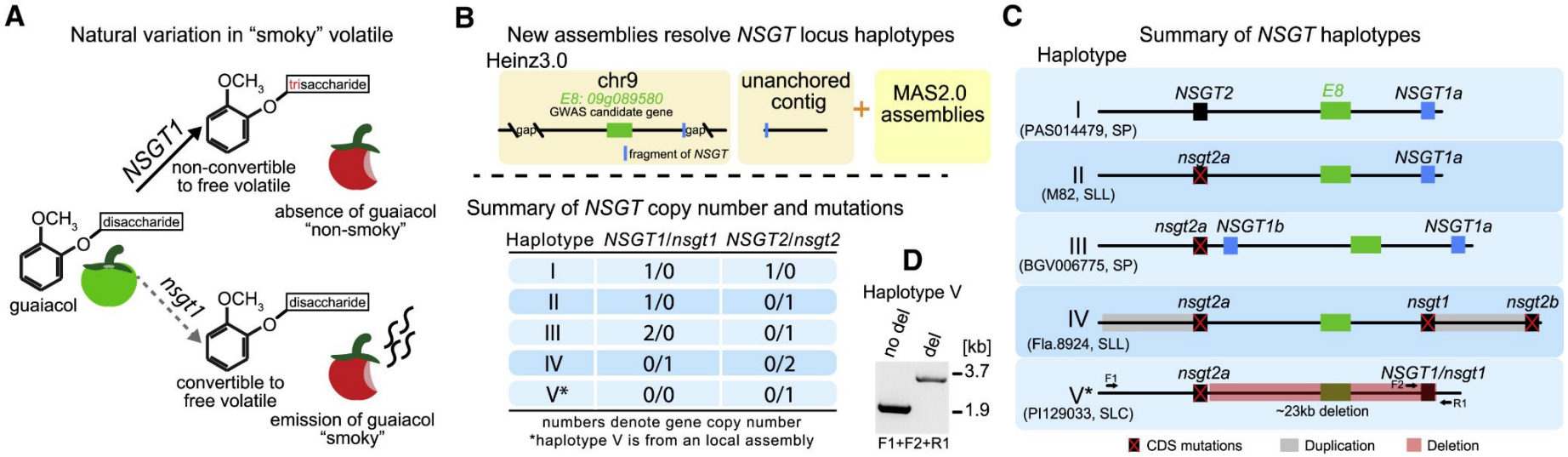


You can generate these types of plots with panacus!

Alonge et al. 2020

Case study 3: 100 tomatoes

Associated SVs with smoky flavor



Alonge et al. 2020

Case studies

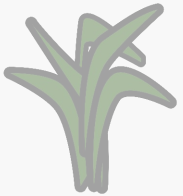
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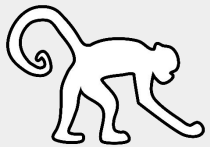
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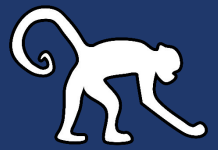


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Case study 4: *Pan* pangenome



New Results

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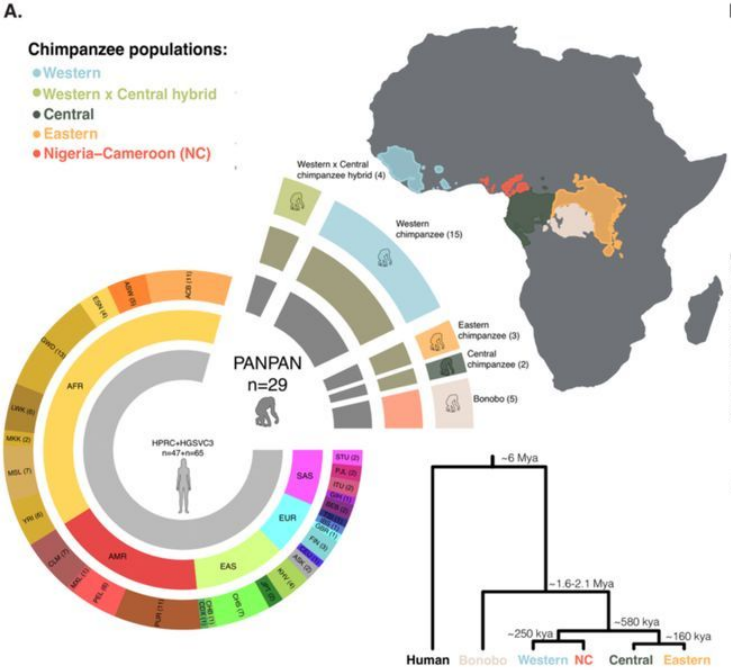
A *Pan*-pangenome illuminates complex structural variation and selection in humans, chimpanzees, and bonobos

Joana L. Rocha, Runyang Nicolas Lou, Carolina Lima Adam, Prajna Hebbar, Scott Ferguson, Davide Bolognini, Alison Killilea, Kendra Hoekzema, Andrea Guarracino, Yun Deng, Nicole Soranzo, Benedict Paten, Erik Garrison, Alex Pollen, Evan E. Eichler, Rori V. Rohlfs, Matthew W. Mitchell, Peter H. Sudmant

doi: https://doi.org/10.64898/2026.06.06.730619

This article is a preprint and has not been certified by peer review [what does this mean?].

Case study 4: *Pan* pangenome

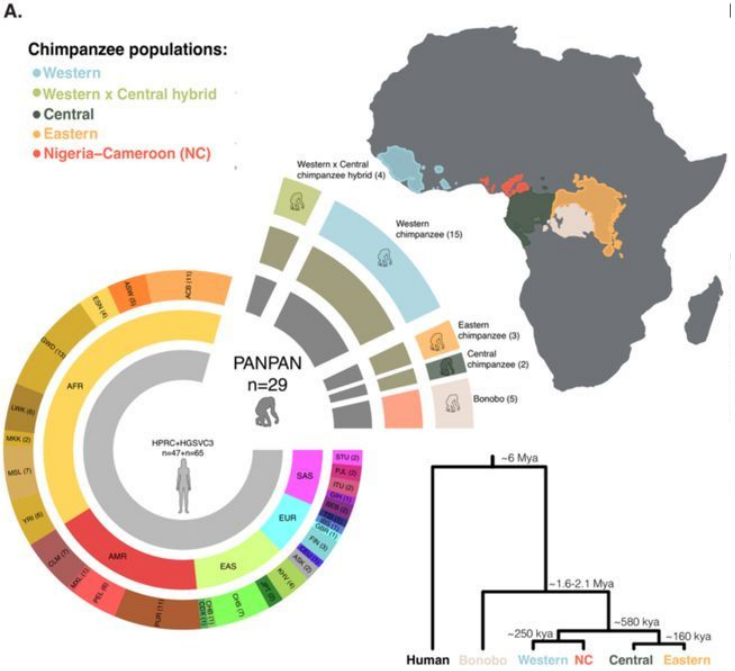


Data:

- 57 haplotypes from 4 *Pan* clades
- HiFi, ONT, and HiC data

Rocha et al. 2026

Case study 4: *Pan* pangenome



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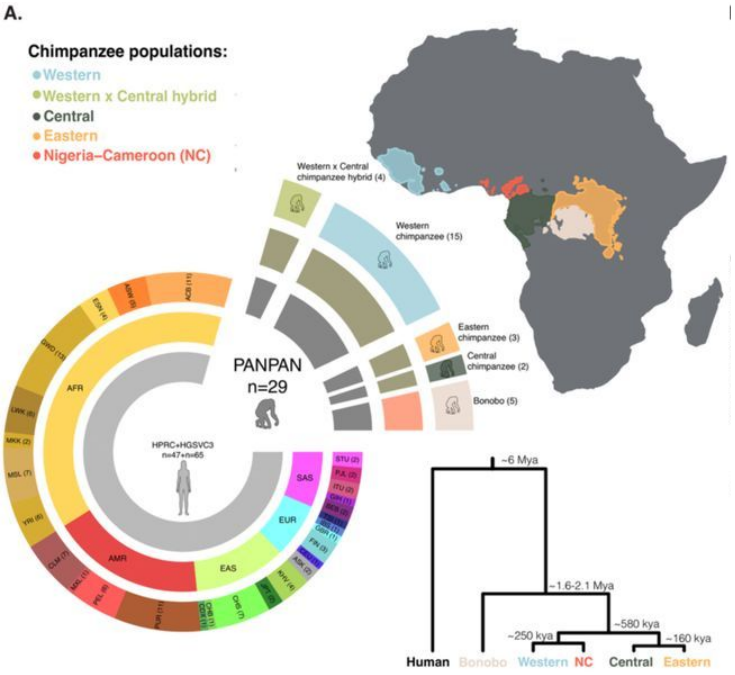
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Rocha et al. 2026

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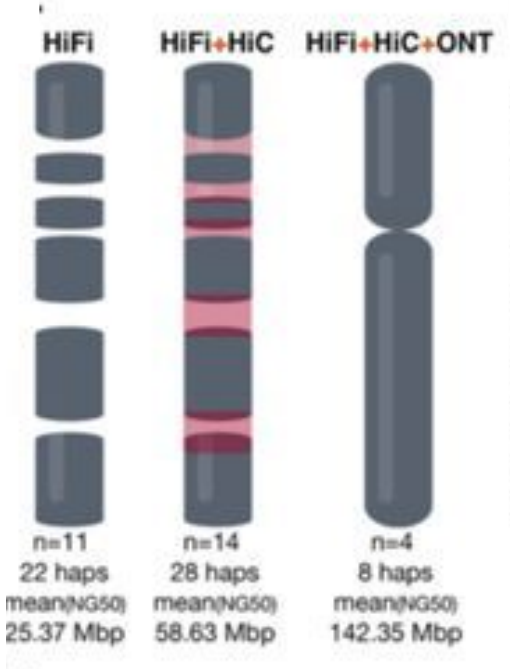
Main results:

- Recent chimp adaptations
- Humans have unique disease causing tandem repeat expansions
- Found evidence of parallel adaptation to malaria in humans and chimps

Rocha et al. 2026

Case study 4: *Pan* pangenome

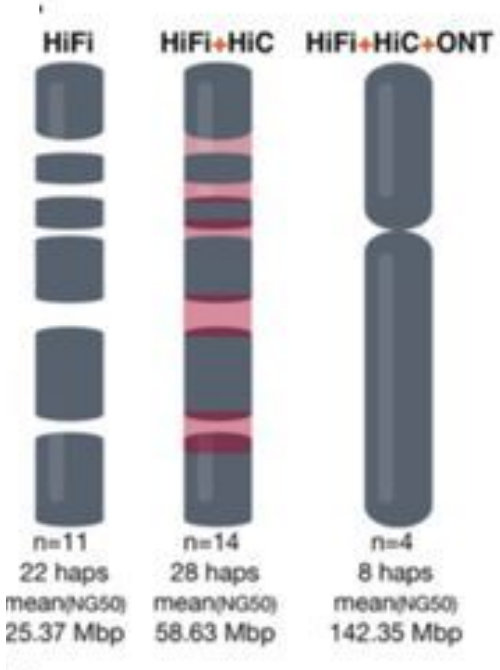
Data and QC



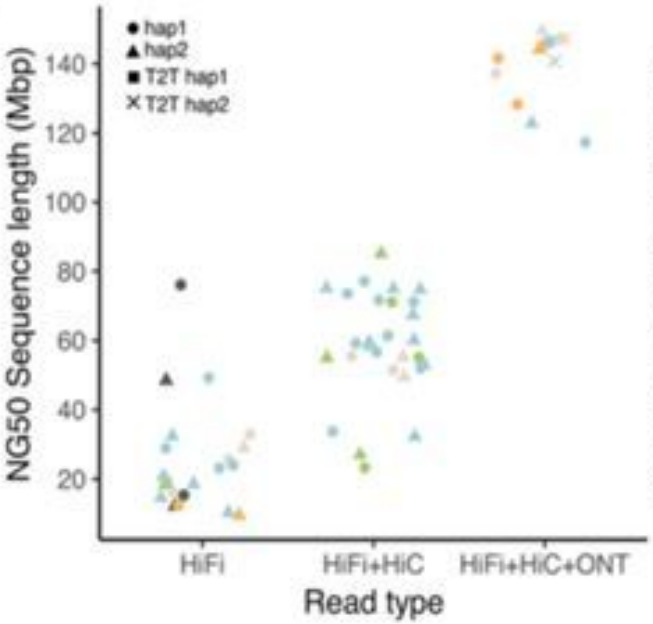
Rocha et al. 2026

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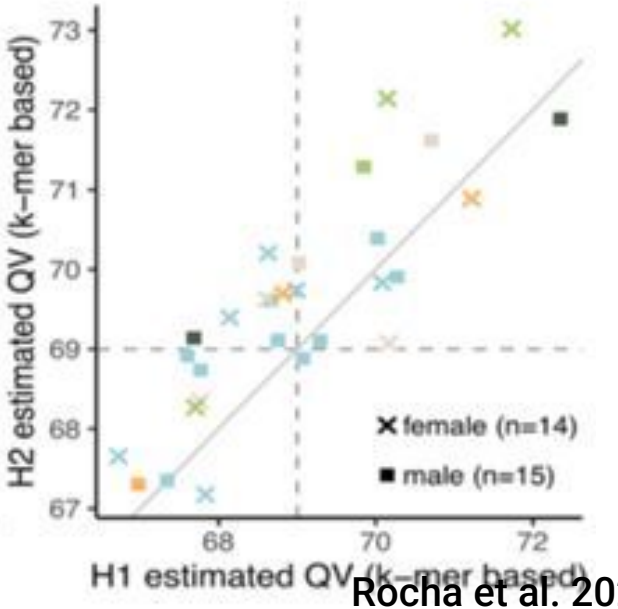
Data and QC



NG50 by data type



QV from Merqury

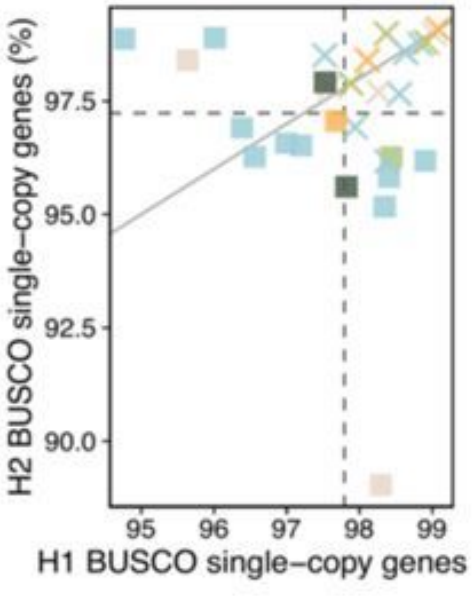


Rocha et al. 2026

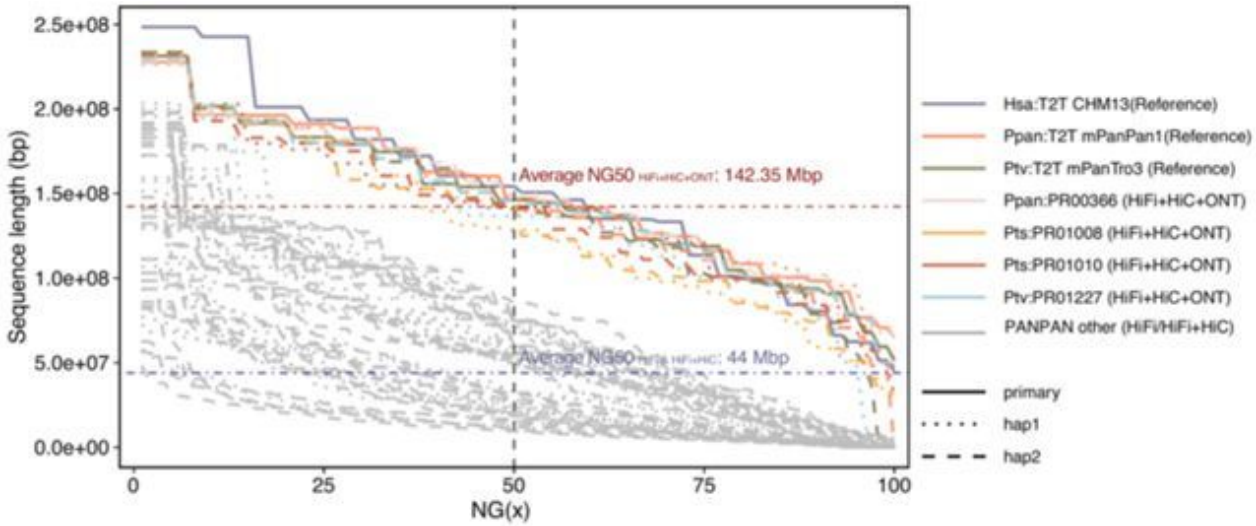
Case study 4: *Pan* pangenome

Genome QC

BUSCOs



NG(x) chart

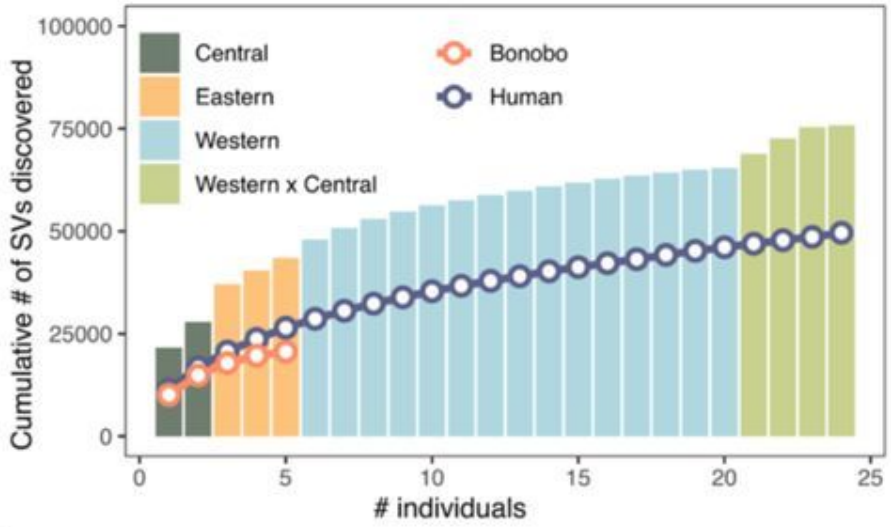


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Case study 4: *Pan* pangenome

Growth curve

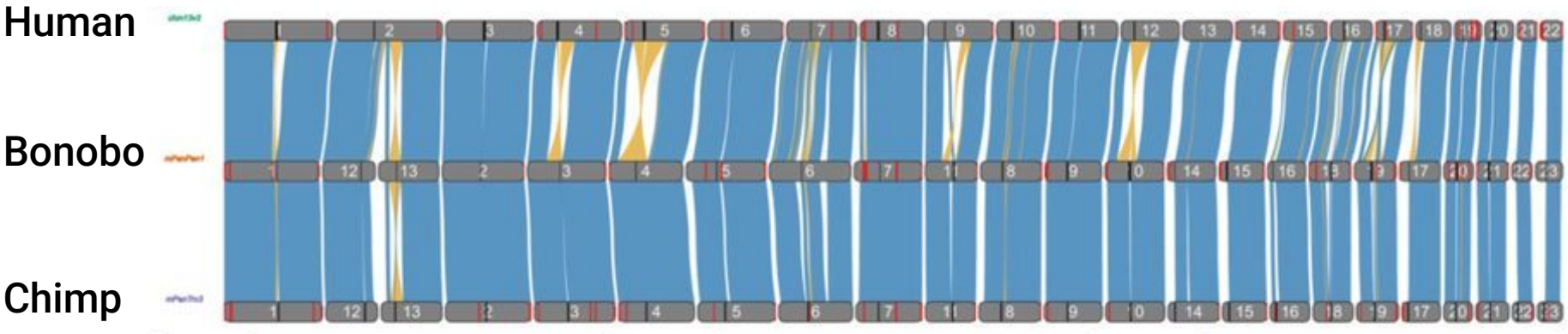
Similar to Panacus



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Case study 4: *Pan* pangenome

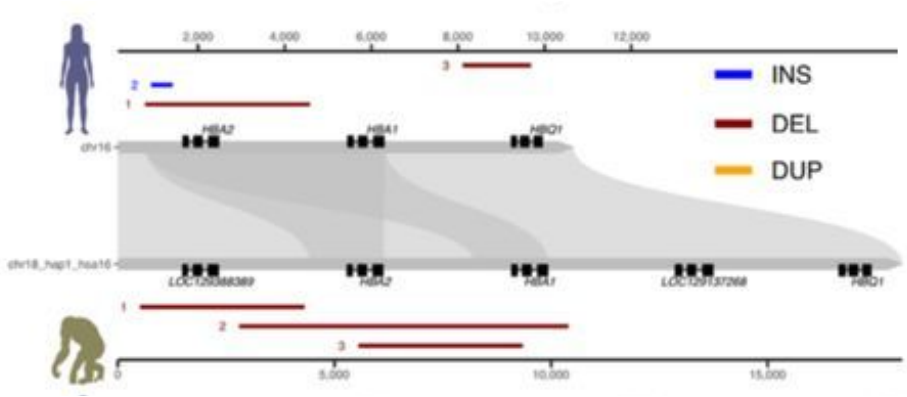
Synteny plot

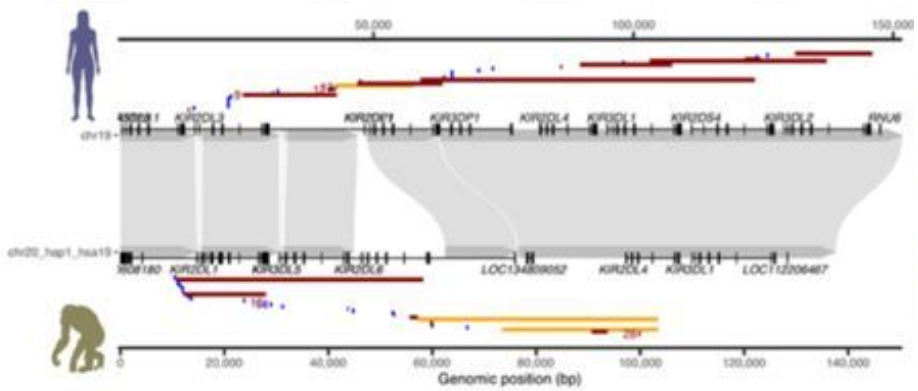


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Case study 4: *Pan* pangenome

Example of functional convergence between humans and chimps





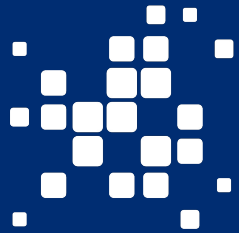
Both Chimps and Humans exhibit extensive deletions/duplications of KIR genes, which are important for malaria resistance

Rocha et al. 2026

Recap: There are a lot of ways to build and use a pangenome

- Start with reads
- QC
- Often (not always) build a genome
- QC
- Compare the genomes / reads
- Use the representation to explore your question of interest

Questions?



wellcome sanger institute

Extras

Case study 5: Humans



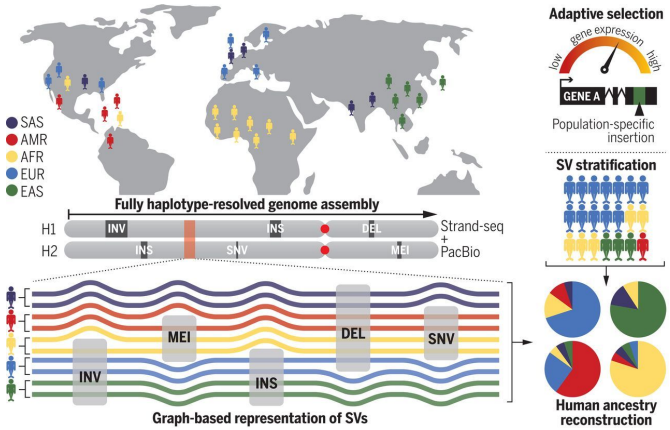
RESEARCH ARTICLE

Haplotype-resolved diverse human genomes and integrated analysis of structural variation

PETER EBERT, PETER A. AUDANO, QIHUI ZHU, BERNARDO RODRIGUEZ-MARTIN, DAVID PORUBSKY, MARC JAN BONDER, ARVIS SULOVARI, JANA EBLER, WEICHEN ZHOU, [...], AND EVAN E. EICHLER +55 authors Authors Info & Affiliations

SCIENCE · 25 Feb 2021 · Vol 372, Issue 6537 · DOI: 10.1126/science.abf7117

Case study 5: Humans

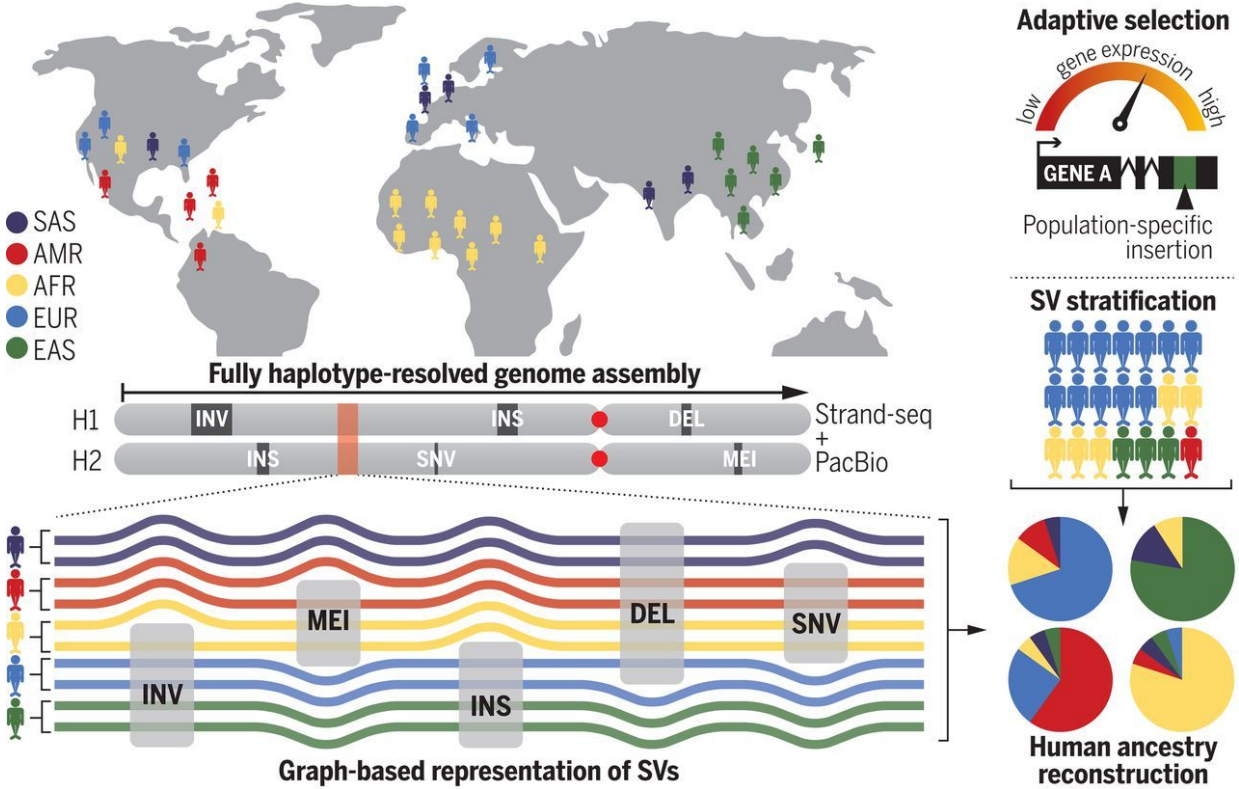


Data:

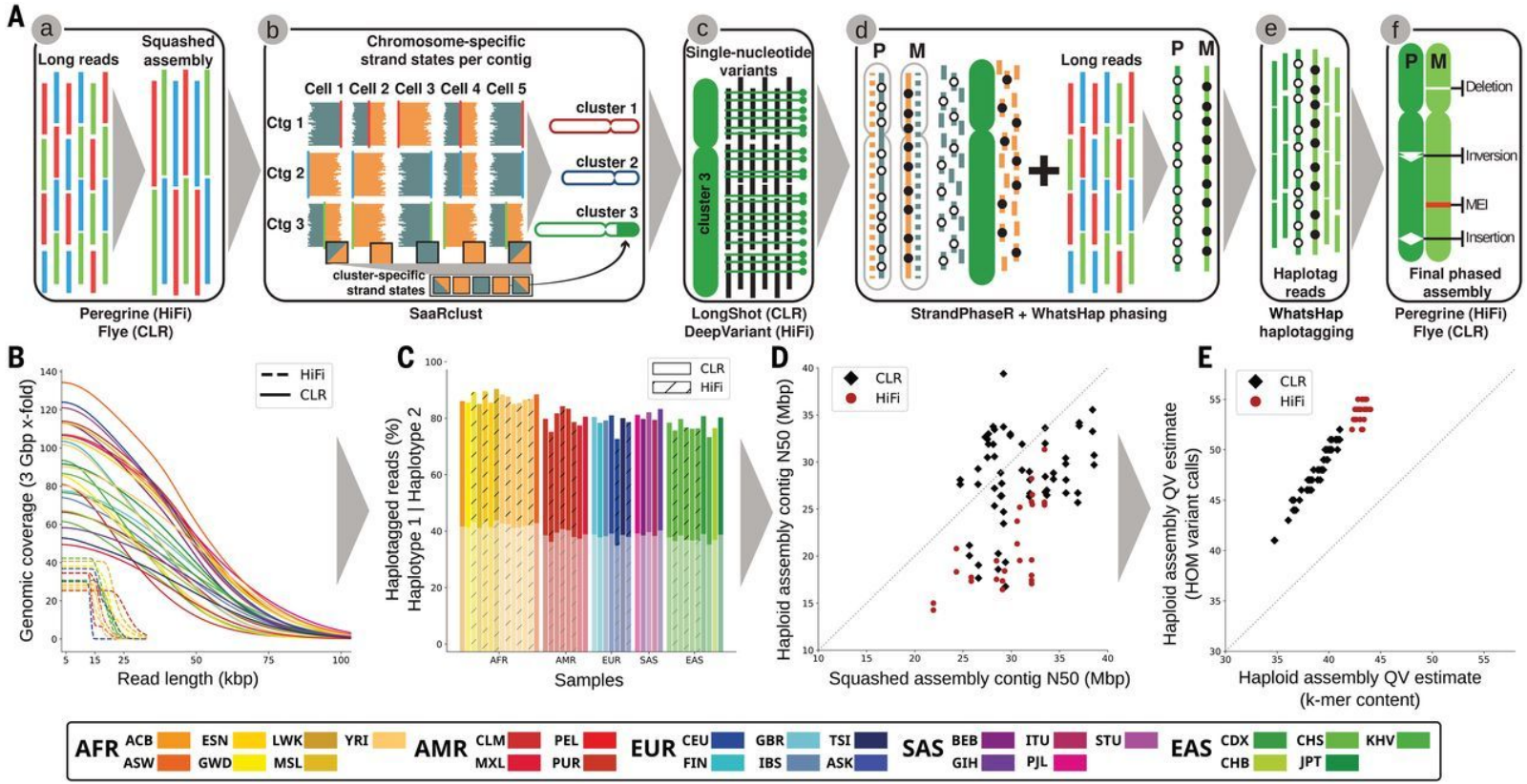
Main results:

Methods:

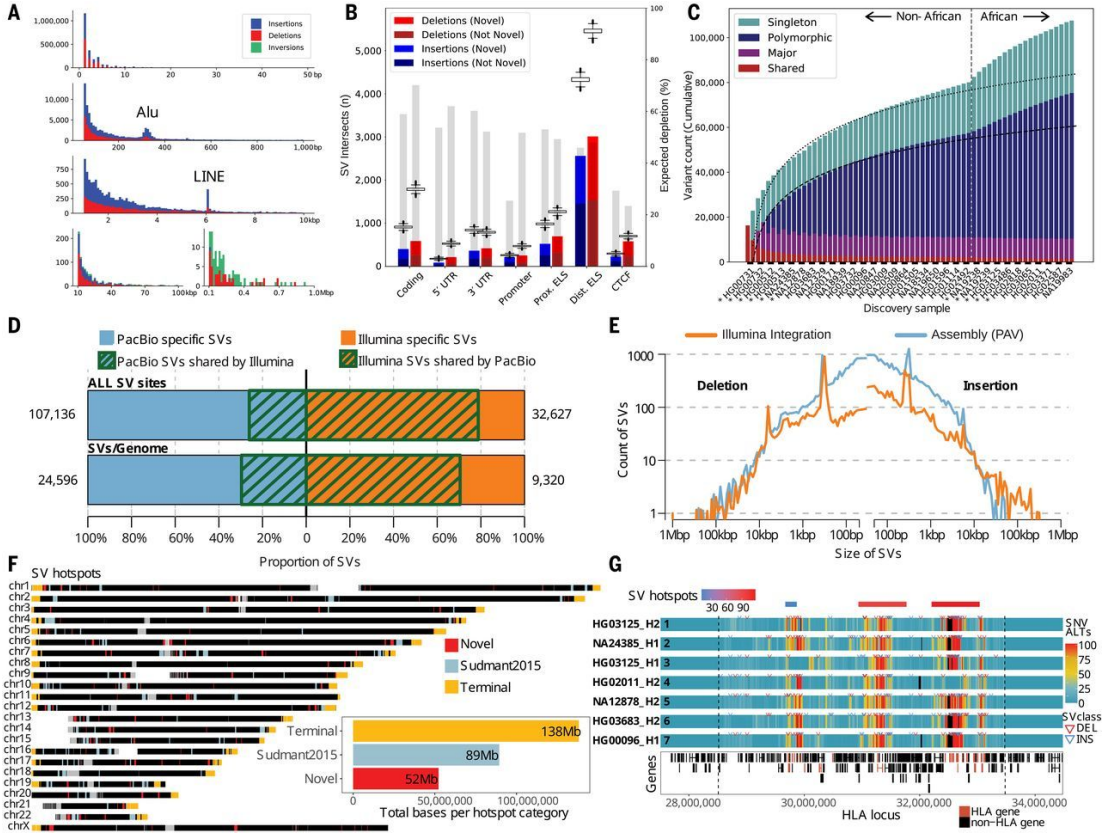
Case study 5: Humans



Case study 5: Humans



Case study 5: Humans



Case study 5: Humans

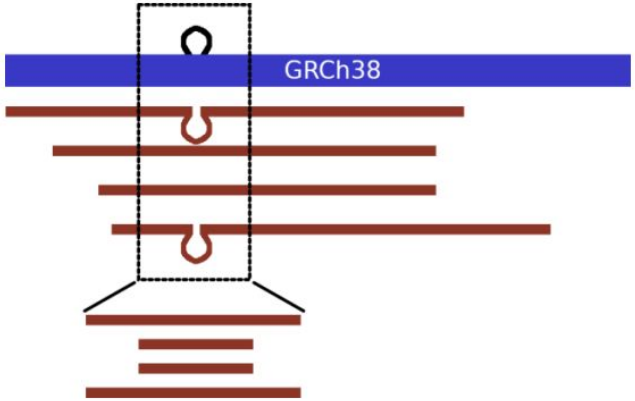


Fig. S70. Subseq illustration

Reads (red) aligned to the reference (blue) with an SV insertion (loop). A region surrounding the breakpoints is chosen (dashed box) with larger regions for larger SVs. All reads traversing that region (i.e., touching both ends) are extracted, and the length of the read in that region is determined by parsing the CIGAR string. In this example, regions extracted from two reads (top and middle) support a heterozygous insertion, and two reads (middle two) do not support it.

Case study 5: Humans

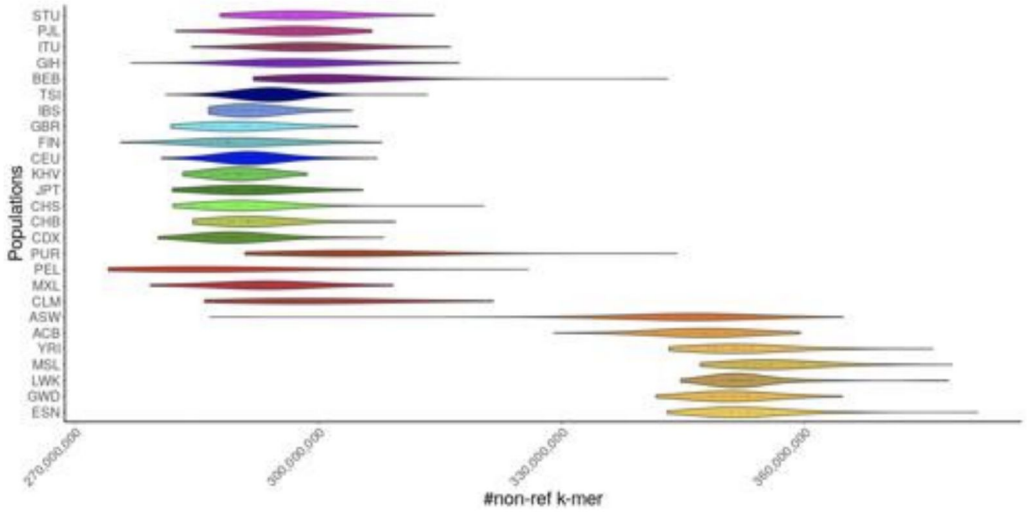


Fig. S87. Non-reference k-mer density distribution by population for 2,504 unrelated samples

Count of non-reference k-mers (k = 61, x-axis) aggregated per population (y-axis, see Fig. 1 for population color legend) for 2,504 unrelated high-coverage Illumina samples.

Genome scaffolding

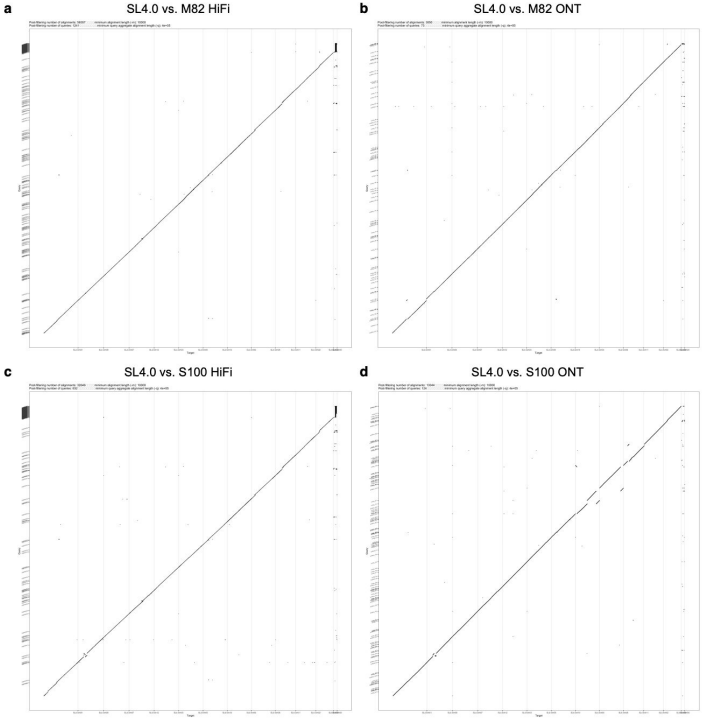
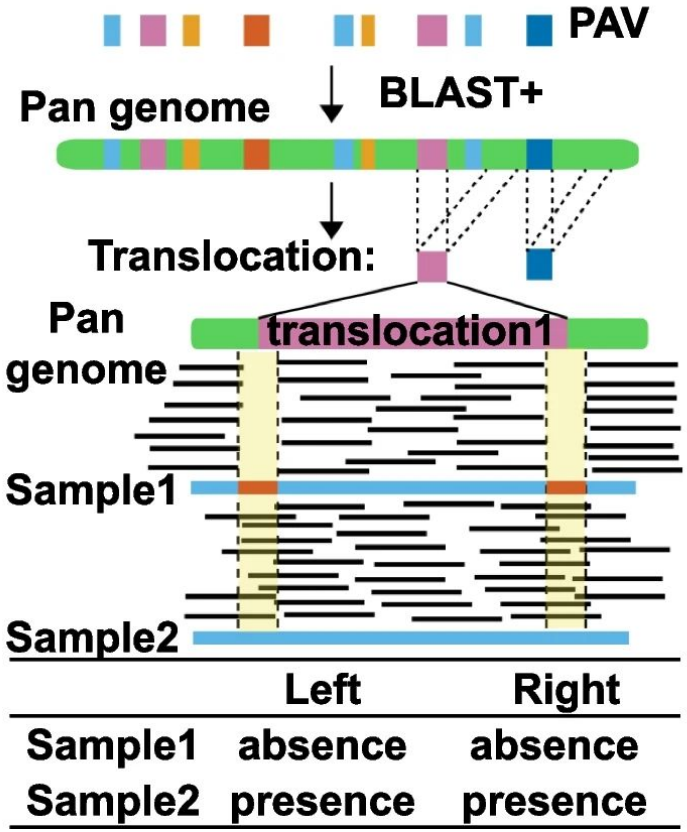
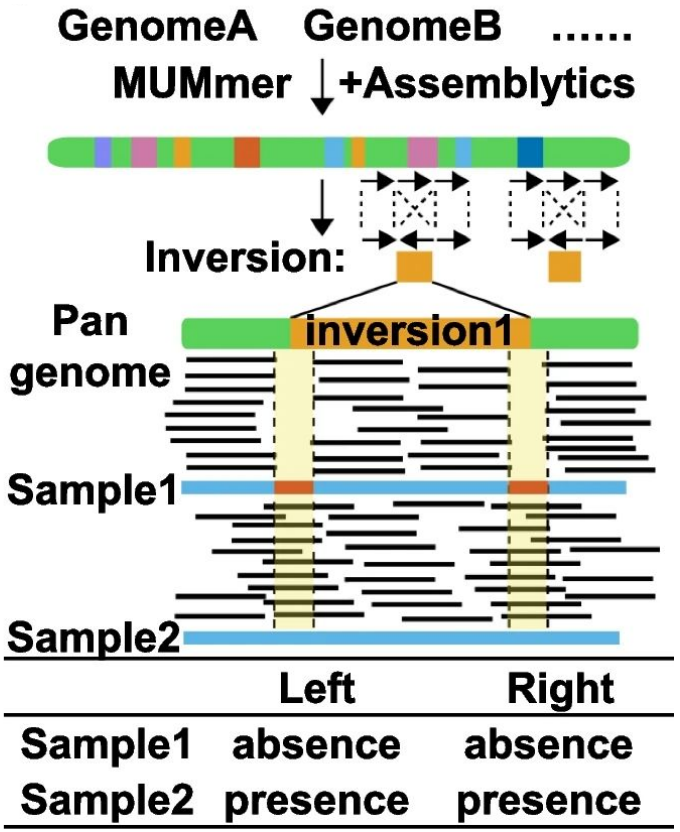


Fig. S11 | Dotplots comparing M82 and S100 draft assemblies to the tomato SL4.0 reference. a and b, Dotplots of M82 HiFi (a) and ONT (b) assemblies versus the tomato SL4.0 reference assembly. **c and d**, Dotplots of Sweet-100 HiFi (c) and ONT (d) assemblies versus the tomato SL4.0 reference assembly.

Case study 2: Rice pangenome GWAS



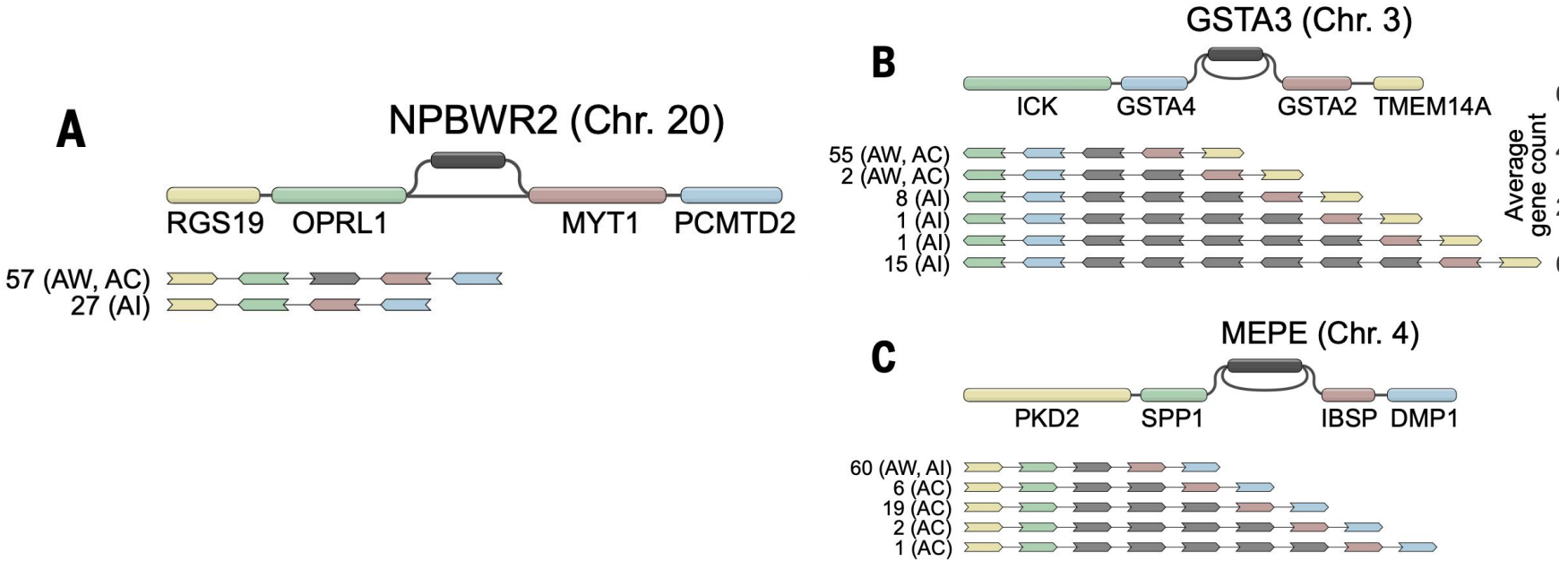
	Left	Right
Sample1	absence	absence
Sample2	presence	presence



	Left	Right
Sample1	absence	absence
Sample2	presence	presence

Case study 1: Scrub jay pangenome

Variants of important bird genes



Edwards et al. 2025